# Agentic IoT: Architectures, Applications, and Challenges Toward the Internet of Agents

# 代理式物聯網：邁向代理人互聯網的架構、應用與挑戰

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## Abstract 摘要

The integration of AI into Internet of Things (AIoT) systems has gradually transformed them from passive data collection infrastructures into intelligent systems capable of anomaly detection, predictive maintenance, classification, forecasting, and optimization. However, most existing solutions still rely on task-specific models that infer from sensor data; thus, system-wide capabilities such as real-time reasoning, adaptive planning, autonomous coordination, learning, tool use, and contextual decision-making remain limited. This paper examines Agentic IoT as a next-generation cognitive IoT paradigm that integrates the perception, reasoning, planning, learning, and action capabilities of autonomous AI agents with cyber-physical systems. Agentic IoT aims to transform IoT from data-centric sensing and inference infrastructures into distributed cognitive agent ecosystems operating across the device/edge-fog-cloud continuum. The paper first grounds this transition as a paradigm shift and positions Agentic IoT in relation to AIoT, edge intelligence, multi-agent systems, and the Internet of Agents. It then systematically reviews current studies, presents a holistic architectural framework, discusses domain-specific application potential, and identifies key technical, operational, and research challenges together with future research directions.

人工智慧與物聯網（AIoT）系統的整合，已逐漸將其從被動的數據收集基礎設施，轉型為具備異常檢測、預測性維護、分類、預測及優化能力的智慧系統。然而，現有的大多數解決方案仍依賴於針對特定任務的感測數據推論模型；因此，諸如即時推理、自適應規劃、自主協調、學習、工具使用及情境決策等系統級能力仍然受限。本文探討「代理式物聯網」（Agentic IoT）作為下一代認知物聯網範式，將自主 AI 代理人的感知、推理、規劃、學習與行動能力與虛實整合系統（CPS）相結合。代理式物聯網旨在將物聯網從以數據為中心的感測與推論基礎設施，轉型為跨越設備/邊緣-霧-雲連續體的分布式認知代理人生態系統。本文首先將此轉型定位為一種範式轉移，並釐清代理式物聯網與 AIoT、邊緣智慧、多代理人系統及代理人互聯網（Internet of Agents）之間的關係。隨後，本文系統性地回顧了當前的研究，提出了一個整體的架構框架，討論了特定領域的應用潛力，並指出了關鍵的技術、營運與研究挑戰，以及未來的研究方向。

Index Terms-Agentic IoT, Internet of Agents, Internet of Things, Autonomous Agents, Large Language Models(LLMs)

索引詞—代理式物聯網 (Agentic IoT)、代理人互聯網 (Internet of Agents)、物聯網 (Internet of Things)、自主代理人 (Autonomous Agents)、大型語言模型 (LLMs)

## I. Introduction Ⅰ. 前言

In recent years, rapid developments in Large Language Models (LLMs), autonomous AI agents, and the design and development of intelligent systems have initiated a fundamental transformation [1]. While traditional machine and deep learning models effectively perform classification, prediction, or optimization tasks, today's agentic AI systems are capable of perceiving contextual information, performing reasoning, creating multi-step plans, utilizing external tools, learning from feedback, and collaborating with other agents to accomplish complex goals [2]. Thanks to these advanced capabilities, a significant transformation is currently taking place from reactive AI systems toward a world of autonomous and goaloriented intelligent agents.

近年來，大型語言模型 (LLMs)、自主 AI 代理人以及智慧系統設計與開發的快速發展，引發了一場根本性的變革 [1]。傳統的機器學習與深度學習模型能有效執行分類、預測或優化任務，而現今的代理式 AI 系統則具備感知上下文資訊、進行推理、制定多步驟計畫、利用外部工具、從回饋中學習，以及與其他代理人協作以達成複雜目標的能力 [2]。得益於這些先進功能，目前正經歷從反應式 AI 系統向自主且目標導向的智慧代理人世界的重大轉型。

The implications of this transformation are particularly significant for the Internet of Things (IoT). As the number of heterogeneous devices connected to IoT increases, the management of heterogeneous devices, dynamic operating conditions, and real-time decision-making requirements have also increased exponentially [3]. Smart-X approaches in many domains, such as smart homes, energy smart environments, smart cities, intelligent transportation networks, and precision

這場變革對物聯網 (IoT) 的影響尤為顯著。隨著連接到物聯網的異質設備數量增加，對於異質設備的管理、動態運行條件以及即時決策的需求也呈指數級增長 [3]。在許多領域的 Smart-X 方法中，例如智慧家庭、能源智慧環境、智慧城市、智慧交通網絡以及精準農業

agriculture applications, increasingly highlight the requirements for distributed decision-making, contextual awareness, real-time coordination, autonomous operation, and adaptation to changing environmental conditions [4]. Therefore, nextgeneration IoT systems are expected not only to collect data and provide communication, but also to understand context, make autonomous decisions, coordinate their actions, and adapt to changing environmental conditions.

農業應用日益強調分散式決策、情境感知、即時協調、自主運作以及對環境變化適應能力的需求 [4]。因此，下一代物聯網（IoT）系統不僅被期望能收集數據並提供通訊，還需具備理解情境、做出自主決策、協調行動並適應不斷變化的環境條件之能力。

依照 Emergence浮現 出現旨在，

In line with these needs in the IoT domain, as in many other fields, new concepts have emerged aiming to enable autonomous agents to operate effectively in large-scale and distributed systems [5]. For example, recent developments in agentic AI have inspired the emergence of a broader vision referred to as the Internet of Agents (IoA) [6]. IoA evolves the traditional Internet paradigm from human-centered communication toward large-scale interaction, coordination, and collaboration among autonomous intelligent agents. The IoA approach places autonomous agents at the center of decisionmaking and computation processes. Within this vision, intelligent agents are envisioned to discover services, negotiate tasks, share knowledge, coordinate actions, and collectively solve complex problems [7].

為了滿足物聯網領域的這些需求，如同許多其他領域一樣，旨在使自主代理（autonomous agents）能在大型分散式系統中有效運作的新概念應運而生 [5]。例如，代理式人工智慧（Agentic AI）的最新發展啟發了一個更廣泛的願景，稱為「代理物聯網」（Internet of Agents, IoA）[6]。IoA 將傳統的網路範式從以人為中心的通訊，演進為自主智慧代理之間的大規模互動、協調與協作。IoA 方法將自主代理置於決策與運算過程的核心。在此願景下，智慧代理被預期能發現服務、協商任務、共享知識、協調行動，並共同解決複雜問題 [7]。

包含

Integrating the conceptual and methodological vision provided by IoA into IoT ecosystems involves unique opportunities and challenges. IoT environments possess characteristics such as resource constraints, distributed infrastructures, heterogeneous devices, intermittent connectivity, real-time requirements, and direct interaction with the physical world [8]. These characteristics require a specialized perspective that integrates the capabilities of autonomous agents with the operational realities of IoT systems. In this direction, we propose the concept of Agentic IoT, which refers to the integration of autonomous agent capabilities with IoT systems, and position it as an application-oriented extension of the Internet of Agents vision in the IoT domain.

以應用為導向的

將 IoA 提供的概念與方法論願景整合到 IoT 生態系統中，既帶來了獨特的機遇，也伴隨著挑戰。IoT 環境具有資源受限、分散式基礎設施、異質設備、間歇性連線、即時性要求以及與物理世界直接互動等特性 [8]。這些特性需要一種專門的視角，將自主代理的能力與 IoT 系統的運作現實相結合。朝此方向，我們提出了「代理式物聯網」（Agentic IoT）的概念，意指將自主代理能力與 IoT 系統整合，並將其定位為 IoA 願景在 IoT 領域中面向應用的延伸。

超越

Within the scope of Agentic IoT, IoT systems evolve beyond mere data collection and monitoring infrastructures into cognitive ecosystems capable of making proactive decisions, adapting to changing environmental conditions, accomplishing goaloriented tasks, and collaborating with other agents [9]. At this point, the task-oriented operating model of AIoT approaches gives way to autonomous, context-aware, and action-oriented systems. Although some studies have emerged in the literature on the Internet of Agents [10], Agentic AI [11], multi-agent systems, and autonomous cyber-physical systems [12], [13], to the best of our knowledge, there is still no comprehensive survey study that integrates these research domains from an

在 Agentic IoT 的範疇內，IoT 系統從單純的數據收集與監控基礎設施，演進為具備認知能力的生態系統，能夠做出主動決策、適應變化的環境條件、完成目標導向的任務，並與其他代理協作 [9]。在此，AIoT 方法的任務導向運作模式，讓位給了自主、情境感知且行動導向的系統。儘管文獻中已出現一些關於代理物聯網 [10]、代理式 AI [11]、多代理系統以及自主虛實整合系統 [12], [13] 的研究，但就我們所知，目前仍缺乏一項全面的綜述研究，能從整合的角度將這些研究領域納入考量。

Table 1: TABLE I 表 1： TABLE I
Comparison of IoT, AIoT, and Agentic IoT Paradigms

IoT、AIoT 與代理型 IoT（Agentic IoT）範式的比較

| Feature 特性 | IoT | AIoT | Agentic IoT 代理式物聯網 (Agentic IoT) |
|---|---|---|---|
| Decision Mechanism 決策機制 | Rule-based/event driven 基於規則/事件驅動 | Model-based (ML/DL) 基於模型 (ML/DL) | LLM-driven reasoning LLM 驅動推理 |
| Autonomy Level 自治程度 | Low 低 | Low/Medium 低/中 | Medium/High 中高 |
| Reasoning Capability 推理能力 | × | Task-spesific 特定任務型 | Explicit reasoning (CoT, ReAct)<br>顯式推理 (CoT, ReAct) |
| Planning 規劃 | Predefined workflows 預定義工作流 | Optimization or RL-based<br>基於最佳化或強化學習 (RL) | Task decomposition, dynamic replanning<br>任務分解、動態重新規劃 |
| Tool Use / API Calling<br>工具使用 / API 呼叫 | System-level API integration<br>系統級 API 整合 | Model-assisted service integration<br>模型輔助服務整合 | Agentic tool use, Function calling<br>代理式工具使用、函式呼叫 |
| Memory 記憶 | Logs, historical records<br>日誌、歷史紀錄 | Historical data, Model states<br>歷史數據、模型狀態 | Short-term + Long-term 短期 + 長期 |
| Learning 學習 | × | Offline/online training, RL<br>離線/線上訓練、強化學習 (RL) | Continuous, Retrieval-augmented learning<br>持續學習、檢索增強學習 |
| Adaptivity 適應性 | Static 靜態 | Limited/partially 有限/部分 | Self-adaptive, Context aware<br>自適應、情境感知 |
| Multi-Agent Coordination<br>多代理人協作 | × | × | Multi-agent collaboration (MCP, A2A)<br>多代理人協作 (MCP, A2A) |
| Natural Language Interaction<br>自然語言互動 | × | × | ✓ |
| Behavior Model 行為模型 | Reactive 反應式 | Predictive 預測式 | Proactive + Goal-driven 主動性 + 目標導向 |

3大比較
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IoT perspective and examines the concept of Agentic IoT in all its aspects. Although each of these research areas contributes to the development of autonomous IoT ecosystems, they often employ different terminologies, assumptions, architectural abstractions, and evaluation methodologies. As a result, the concept of Agentic IoT has not yet been clearly defined as a holistic and systematic research paradigm, and its relationship with related concepts has not been sufficiently clarified.

物聯網（IoT）視角，並全面檢視代理式物聯網（Agentic IoT）的概念。儘管這些研究領域各自對自主物聯網生態系統的發展有所貢獻，但它們往往採用不同的術語、假設、架構抽象化和評估方法。因此，代理式物聯網的概念尚未被明確定義為一個整體且系統化的研究範式，其與相關概念之間的關係也尚未得到充分釐清。

Motivated by the aforementioned research gaps, this study aims to systematically address the concept of Agentic IoT and establish a holistic foundation for this emerging research field. The main contributions of this study are summarized as follows:

基於上述研究缺口，本研究旨在系統性地探討代理式物聯網的概念，並為這一新興研究領域建立整體的基礎。本研究的主要貢獻總結如下：

- We formally define the concept of Agentic IoT and explain its relationship with IoA, AIoT, and Agentic AI.

  - 我們正式定義了代理式物聯網的概念，並解釋其與代理人物聯網（IoA）、人工智慧物聯網（AIoT）及代理式人工智慧（Agentic AI）之間的關係。

- We present the paradigm shift from traditional IoT and AIoT systems toward autonomous, agent-centric, and cognitive IoT ecosystems.

- 我們呈現了從傳統物聯網（IoT）與人工智慧物聯網（AIoT）系統，轉向自主、以代理人為中心且具備認知能力的物聯網生態系統之範式轉移。

- We comprehensively review the current literature, enabling technologies, reference architectures, communication mechanisms, and prominent application domains of Agentic IoT.

  - 我們全面回顧了代理型物聯網（Agentic IoT）的現有文獻、致能技術、參考架構、通訊機制以及主要的應用領域。

- We identify the major challenges and future research directions related to the real-world deployment of Agentic IoT systems.

  - 我們識別了與代理型物聯網系統在現實世界部署相關的主要挑戰與未來研究方向。

The remainder of this paper is organized as follows. Section II discusses the evolution from traditional IoT systems to Agentic IoT and the fundamental paradigm shifts underlying this transformation. Section III presents the proposed Agentic IoT architecture, including the reference architecture, agent cognitive loop, memory and tool integration mechanisms, communication frameworks and the deployment strategies. Section IV examines Agentic IoT applications across different domains. Section V discusses the key challenges and outlines future research directions. Finally, Section VI concludes the paper.

本文其餘部分的結構安排如下：第二節討論從傳統物聯網系統到代理型物聯網的演進，以及支撐此一轉型的根本範式轉移。第三節介紹所提出的代理型物聯網架構，包括參考架構、代理人認知迴圈、記憶與工具整合機制、通訊框架以及部署策略。第四節檢視代理型物聯網在不同領域的應用。第五節討論關鍵挑戰並概述未來研究方向。最後，第六節對全文進行總結。

## II. From Traditional IoT to Agentic IoT: A Paradigm Shift

## II. 從傳統物聯網到代理式物聯網：典範轉移

In recent years, due to rapid and emerging developments in the field of AI, significant paradigm shifts have been experienced in the Internet of Things domain, as in many other fields and technologies. Traditional IoT systems were largely focused on collecting data from the physical environment and processing this data through event-driven mechanisms on cloud-centric infrastructures. In the following stage, computational capabilities were moved toward fog and edge networks, enabling real-time analytics and low-latency decision-making mechanisms [14]. However, existing fog and edge computing approaches still exhibit reactive behavioral models running on static learning pipelines that are unable to inherently reason about environmental uncertainties, long-term objectives, causal relationships, and dynamic operational constraints. During this period, the concept of AIoT (Artificial Intelligence of Things) emerged through the integration of ML and AI algorithms into IoT systems to address existing challenges [15]. Although AIoT is still used in many domains and systems, the models employed in these systems inherently operate on pre-trained static models. Therefore, they remain inadequate in situations that require heterogeneous, dynamic, uncertain, and real-time reasoning capabilities. These limitations have clearly revealed the need for a paradigm in IoT systems that can provide contextual understanding, autonomous decision-making, and adaptive behavior [9].

近年來，由於人工智慧領域的快速與新興發展，物聯網領域與許多其他領域和技術一樣，經歷了重大的典範轉移。傳統的物聯網系統主要集中於從物理環境收集數據，並透過以雲端為中心的基礎設施，利用事件驅動機制來處理這些數據。在隨後的階段，運算能力向霧運算與邊緣網路遷移，實現了即時分析與低延遲的決策機制 [14]。然而，現有的霧運算與邊緣運算方法仍表現出運行在靜態學習管線上的反應式行為模型，無法對環境的不確定性、長期目標、因果關係以及動態操作約束進行內在推理。在此期間，透過將機器學習（ML）與人工智慧（AI）演算法整合到物聯網系統中，產生了 AIoT（人工智慧物聯網）的概念，以應對現有的挑戰 [15]。儘管 AIoT 仍被廣泛應用於許多領域和系統，但這些系統所採用的模型本質上是基於預訓練的靜態模型運作。因此，在需要異質、動態、不確定且即時推理能力的場景中，它們仍然顯得不足。這些限制清楚地揭示了物聯網系統需要一種能夠提供情境理解、自主決策和自適應行為的新典範 [9]。

### Definition: Agentic IoT 定義：代理式物聯網 (Agentic IoT)

Agentic IoT is a next-generation cognitive IoT paradigm that transforms IoT from networks of connected devices into distributed cognitive agent ecosystems by integrating the perception, reasoning, planning, learning, and action capabilities of autonomous AI agents with cyber-physical systems.

代理式物聯網（Agentic IoT）是下一代認知物聯網典範，它透過將自主 AI 代理的感知、推理、規劃、學習與行動能力與虛實整合系統（CPS）相結合，將物聯網從連接裝置的網路轉型為分散式的認知代理生態系統。

With the development of LLMs, AI systems have gained advanced capabilities such as reasoning, planning, tool use,

隨著 LLMs 的發展，AI 系統已獲得了如推理、規劃、工具使用、

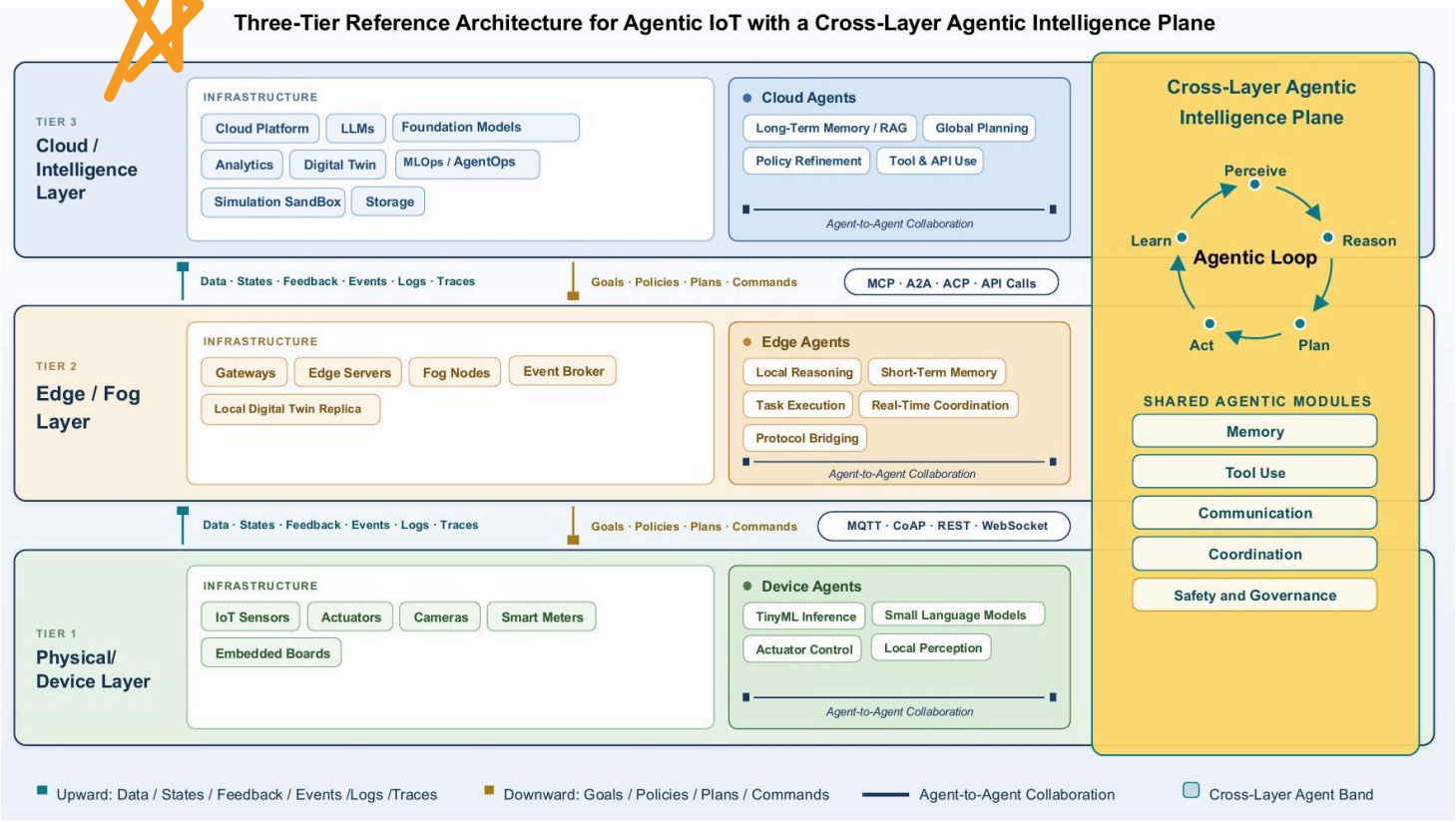



Figure 1: Fig. 1. Three-Tier Reference Architecture for Agentic IoT

圖 1：代理式物聯網（Agentic IoT）的三層參考架構

multi-step decision-making, and the interpretation of environmental context [16]. This evolution represents a paradigm shift in IoT research, where connected devices are no longer viewed merely as passive sensing and actuation endpoints, but as components of autonomous, context-aware, and goal-directed cyber-physical ecosystems. In this study, we use the term Agentic IoT to denote this emerging paradigm, as formalized in the definition box.

多步決策以及環境上下文解讀等進階能力 [16]。這種演進代表了物聯網研究的範式轉移，聯網設備不再僅被視為被動的感測與執行端點，而是作為自主、具備上下文感知且目標導向的虛實整合生態系統之組成部分。在本研究中，我們使用「代理式物聯網」（Agentic IoT）一詞來表示這一新興範式，如定義框中所正式定義。

## III. Agentic IoT Architecture

## III. 代理式物聯網架構

This section describes how the Agentic IoT vision can be realized as a concrete system design. The guiding principle is that agent intelligence should be distributed across the IoT stack in proportion to the resources available at each layer, while all agents share a common cognitive model and a common set of support modules.

本節說明如何將代理式物聯網（Agentic IoT）願景落實為具體的系統設計。其指導原則是：代理程式的智慧應根據各層可用的資源，按比例分布在物聯網堆疊中，同時所有代理程式共享通用的認知模型與一組通用的支援模組。

### A. Reference Architecture

### A. 參考架構

Fig. 1 illustrates the proposed three-tier reference architecture for Agentic IoT. The architecture organizes the system into a physical/device layer, an edge/fog layer, and a cloud/intelligence layer, and complements them with a crosslayer agentic intelligence plane that spans all three. Each tier combines conventional IoT infrastructure with a class of agents whose capabilities are matched to the computational resources available at that level.

圖 1 展示了所提出的代理式物聯網三層參考架構。該架構將系統組織為物理/設備層、邊緣/霧層以及雲端/智慧層，並輔以一個橫跨這三層的跨層代理智慧平面。每一層都將傳統的物聯網基礎設施與一類代理程式相結合，這些代理程式的能力與該層級可用的運算資源相匹配。

The physical/device layer (Tier 1) contains the components that interact directly with the environment: IoT sensors, actuators, cameras, smart meters, and embedded boards. Device agents at this level operate under strict memory, computation, and energy budgets. They therefore rely on TinyML inference and small language models (Table II) for local

perception, and on actuator control modules to turn decisions into physical actions. Their role is deliberately narrow: fast perceptionaction behavior at the point where data is produced, without waiting for the upper layers.

物理/設備層（第一層）包含直接與環境互動的組件：物聯網感測器、致動器、攝影機、智慧電表和嵌入式板卡。此層級的設備代理程式在嚴格的記憶體、運算和能源預算下運作。因此，它們依賴 TinyML 推論和小語言模型（表 II）進行局部感知，並依靠致動器控制模組將決策轉化為物理行動。它們的角色被刻意設定得較為狹窄：在數據產生的端點進行快速的感知-行動行為，而無需等待上層處理。

The edge/fog layer (Tier 2) consists of gateways, edge servers, and fog nodes, supported by an event broker and a local replica of the digital twin. Edge agents reason over data aggregated from many devices, maintain short-term memory of the current operational context, execute tasks, and coordinate devices in real time. This tier also hosts protocol bridging, the mechanism that connects device-level IoT protocols with agent-level communication (Section IIID). Since edge agents sit close to the field but command more resources than any single device, they are the natural place for time-critical decisions that involve multiple devices.

邊緣/霧層（第 2 層）由閘道器、邊緣伺服器和霧節點組成，並由事件代理程式（event broker）和數位分身的本地副本提供支援。邊緣代理程式對來自多個設備的聚合數據進行推理，維護當前操作情境的短期記憶，執行任務，並即時協調設備。此層還承載了協定橋接，即連接設備級 IoT 協定與代理程式級通訊的機制（第 IIID 節）。由於邊緣代理程式靠近現場，但比任何單一設備擁有更多資源，因此它們是涉及多個設備且具時間關鍵性決策的自然首選位置。

Retrieval-Augmented Generation（檢索增強生成）**

The cloud/intelligence layer (Tier 3) provides the heaviest computational resources: cloud platforms, LLMs and foundation models, storage, analytics, digital twins, MLOps/AgentOps pipelines, and a simulation sandbox. Cloud agents use this infrastructure for the capabilities that lower tiers cannot afford: long-term memory with retrieval-augmented generation (RAG), global planning across the whole system, refinement of the policies that govern lower-tier agents, and access to

雲端/智慧層（第 3 層）提供最強大的運算資源：雲端平台、LLMs 和基礎模型、儲存、分析、數位分身、MLOps/AgentOps 流水線以及模擬沙盒。雲端代理程式利用這些基礎設施來實現低層級無法負擔的功能：具備檢索增強生成（RAG）的長期記憶、跨整個系統的全域規劃、精煉管理低層級代理程式的策略，以及存取

external tools and APIs. The simulation sandbox and the digital twin also allow candidate plans to be tested before they are released to the physical system.

外部工具和 API。模擬沙盒和數位分身還允許在候選方案發布到實體系統之前對其進行測試。

The tiers are linked by two vertical information flows moving in opposite directions. Data, states, feedback, events, logs, and traces flow upward, giving higher tiers a progressively wider view of the system. Goals, policies, plans, and commands flow downward, translating high-level objectives into increasingly concrete instructions. Alongside these vertical flows, each tier supports horizontal agent-to-agent collaboration, so peer agents can share tasks and state without routing every interaction through a higher layer.

這些層級透過兩個方向相反的垂直資訊流相連。數據、狀態、回饋、事件、日誌和追蹤向上流動，使更高層級能逐漸獲得更廣泛的系統視野。目標、策略、計畫和指令向下流動，將高階目標轉化為日益具體的指令。除了這些垂直流動外，每個層級都支援水平的代理程式間協作，因此同級代理程式可以共享任務和狀態，而無需將每次互動都路由到更高層級。

What distinguishes this design from conventional layered IoT architectures is the cross-layer agentic intelligence plane shown on the right of Fig. 1. The plane provides two elements that all agents share regardless of where they run. The first is the agentic loop (perceive, reason, plan, act, learn), which defines a common behavioral model for every agent in the system and is examined in Section IIIB. The second is a set of shared agentic modules for memory, tool use, communication, coordination, and safety and governance, implemented once and reused across tiers rather than re-created inside each layer. Placing safety and governance in this shared plane subjects every autonomous action to the same constraints, whether it is a single actuator command at the device level or a global policy update in the cloud. Intelligence, in this architecture, is not the property of one “smart” layer; it is a system-wide capability scaled to the resources of each tier.

這種設計與傳統分層物聯網（IoT）架構的區別在於圖 1 右側所示的跨層代理智慧平面（cross-layer agentic intelligence plane）。該平面提供了所有代理程式（agent）無論在何處運行都共享的兩個要素。第一個是代理迴圈（感知、推理、規劃、行動、學習），它為系統中的每個代理程式定義了通用的行為模型，並在第 III-B 節中進行探討。第二個是一組用於記憶、工具使用、通訊、協調以及安全與治理的共享代理模組，這些模組僅需實作一次即可跨層級重複使用，而非在每個層級內部重新建立。將安全與治理置於此共享平面中，可使每一項自主行動都受到相同的約束，無論是裝置層級的單一執行器指令，還是雲端中的全域策略更新。在這種架構中，智慧並非某個「智慧」層的特性，而是一種根據各層級資源進行縮放的系統級能力。

## B. Agent Cognitive Loop

## B. 代理認知迴圈

At the core of Agentic IoT lies a structure that enables agents to move beyond simple rule execution and presents the processes of sensing, reasoning, planning, acting, and learning within a holistic cognitive loop, shown as the Agentic Loop in Fig. 1.

代理物聯網（Agentic IoT）的核心在於一種使代理程式能夠超越簡單規則執行，並在一個整體的認知迴圈中呈現感測、推理、規劃、行動和學習過程的結構，如圖 1 中的代理迴圈（Agentic Loop）所示。

The loop begins with perception. The agent collects raw observations, including sensor readings, system states, events, and messages from other agents, and converts them into a structured context. In the reasoning stage, the agent interprets this context: it determines what is happening, why it is happening, and what it implies for its current goals. Unlike a threshold rule, this stage can weigh several pieces of evidence together and may employ explicit reasoning strategies such as chainof-thought or ReAct (Table I). Planning then turns conclusions into a course of action. The agent decomposes its goal into steps, selects a tool, an actuator, or a collaborating agent for each step, and revises the plan when conditions change. In the acting stage, the plan is executed: devices are actuated, APIs and tools are called, and subtasks are delegated to other agents. Learning closes the cycle. The agent compares outcomes with its expectations, stores the experience in memory, and adjusts its future reasoning and planning accordingly.

該迴圈始於感知。代理程式收集原始觀測數據，包括感測器讀數、系統狀態、事件以及來自其他代理程式的訊息，並將其轉換為結構化的上下文。在推理階段，代理程式會解讀此上下文：確定正在發生什麼、發生的原因，以及這對其當前目標意味著什麼。與閾值規則不同，此階段可以綜合權衡多項證據，並可能採用明確的推理策略，如思維鏈（chain-of-thought）或 ReAct（表 I）。接著，規劃階段將結論轉化為行動方案。代理程式將其目標分解為多個步驟，為每個步驟選擇工具、執行器或協作代理程式，並在條件變化時修正計畫。在行動階段，計畫被執行：驅動裝置、調用 API 和工具，並將子任務委派給其他代理程式。學習階段則完成了這個循環。代理程式將結果與預期進行比較，將經驗儲存在記憶中，並據此調整未來的推理與規劃。

Two properties separate this loop from the control loops of traditional IoT systems. First, it is stateful: each iteration is informed by the memory of earlier ones, so behavior can improve over time instead of repeating a fixed response. Second, the loop runs at a different depth and timescale in each tier. A device agent completes fast, shallow iterations within milliseconds; an edge agent iterates over seconds while coordinating local tasks; a cloud agent may take minutes or hours for global planning and policy refinement. Because every tier implements the same loop, these cycles nest naturally: the outcome of one slow cloud-level iteration becomes the set of goals and policies within which the many faster iterations below it operate.

這兩個特性將此迴圈與傳統物聯網（IoT）系統的控制迴圈區分開來。首先，它是具備狀態的（stateful）：每一次迭代都會參考先前迭代的記憶，因此行為能隨時間改進，而非重複固定的反應。其次，該迴圈在各個層級中以不同的深度與時間尺度運行。裝置代理（device agent）在毫秒內完成快速且淺層的迭代；邊緣代理（edge agent）在協調局部任務時以秒為單位進行迭代；雲端代理（cloud agent）則可能花費數分鐘或數小時進行全域規劃與策略優化。由於每個層級都執行相同的迴圈，這些週期會自然地嵌套：一次緩慢的雲端層級迭代結果，會成為其下方眾多快速迭代運作時的目標與策略框架。

3k agent

## C. Memory, RAG and Tool Integration

## C. 記憶、RAG 與工具整合

A cognitive loop is only as capable as the state it can keep and the actions it can take. In the proposed architecture, both concerns are handled by shared modules of the agentic intelligence plane (Fig. 1) rather than being left as internal details of individual models.

認知迴圈的能力取決於其能維持的狀態以及能採取的行動。在本文提出的架構中，這兩項事務皆由代理智慧層（Agentic Intelligence Plane，如圖 1 所示）的共享模組處理，而非留作個別模型的內部細節。

Memory is organized in two forms that mirror the tier structure. Short-term memory, held by edge and device agents, stores the immediate operational context: recent observations, the state of the task in progress, and the latest instructions received from above. Long-term memory, maintained in the cloud, accumulates episodic and semantic knowledge over the system's lifetime: past incidents and their resolutions, device behavior profiles, seasonal patterns, and refined policies. Together, they allow an agent to interpret a situation not only through what it currently observes but also through what the system has observed before.

記憶以兩種形式組織，並與層級結構相對應。由邊緣和裝置代理持有的「短期記憶」儲存了即時的運作情境：包含近期的觀測結果、進行中的任務狀態，以及從上層接收到的最新指令。「長期記憶」則維護於雲端，累積系統生命週期中的情節性與語義性知識：包含過去的事件及其解決方案、裝置行為剖析、季節性模式以及優化後的策略。兩者結合，使代理不僅能透過當前的觀測來解讀現況，還能參考系統過往的觀測經驗。

RAG connects this long-term memory to the reasoning stage. Instead of expecting a language model to carry every relevant fact in its parameters, the agent retrieves the pertinent knowledge at decision time (device manuals, historical logs, maintenance records, domain documents) and places it in the model's context. This grounds decisions in verifiable, systemspecific sources, reduces the risk of hallucinated outputs, keeps knowledge updatable without retraining, and lets even the small models of the lower tiers act with information well beyond their parameter capacity.

RAG 將這種長期記憶與推理階段連結起來。代理程式不再期望語言模型在其參數中攜帶所有相關事實，而是在決策時檢索相關知識（設備手冊、歷史日誌、維護記錄、領域文件）並將其放入模型的上下文（context）中。這使決策能以可驗證且特定於系統的來源為依據，降低了生成幻覺輸出的風險，使知識無需重新訓練即可保持更新，並讓底層的微型模型也能運用遠超其參數容量的資訊來行動。

Tool integration plays the complementary role: where memory and retrieval extend what an agent knows, tools extend what it can do. Through function calling, an agent invokes external capabilities as steps of its plan, such as querying a database, requesting a forecast, running a what-if scenario in the simulation sandbox, or issuing a command through a device API. Tool use in Agentic IoT, however, has a property that purely digital agent systems lack: some tool calls produce physical effects. For this reason, tool invocations pass through the shared safety and governance module, which validates them against operational constraints before execution.

工具整合扮演了互補的角色：記憶與檢索擴展了代理程式「知道」的內容，而工具則擴展了它「能做」的事情。透過函式呼叫（function calling），代理程式將調用外部功能作為其計畫的步驟，例如查詢資料庫、請求預測、在模擬沙盒中執行假設分析（what-if scenario），或透過設備 API 發送指令。然而，代理式物聯網（Agentic IoT）中的工具使用具有純數位代理系統所缺乏的特性：某些工具調用會產生物理效應。因此，工具調用必須通過共享的安全與治理模組，在執行前根據操作限制進行驗證。

D. Communication: MCP, A2A, and IoT Protocol Bridging

D. 通訊：MCP、A2A 與物聯網協定橋接

Communication in Agentic IoT spans two protocol families that evolved independently. Established IoT protocols such as MQTT, CoAP, REST, and WebSocket were designed for constrained devices; they carry telemetry and commands efficiently but encode no semantics for goals, capabilities, or negotiation. Emerging agent protocols supply exactly this

代理式物聯網中的通訊跨越了兩個獨立演進的協定家族。既有的物聯網協定（如 MQTT、CoAP、REST 和 WebSocket）是為受限設備設計的：它們能高效地傳輸遙測數據和指令，但並未針對目標、能力或協商進行語義編碼。新興的代理協定則恰好提供了這些功能

Table 2: TABLE II 表 2：TABLE II
(Small) Language Models for Agentic IoT: A Comparative Overview

代理式物聯網（Agentic IoT）之（小）語言模型：比較概覽

| Model 模型 | Params 參數 | Architecture 架構 | Deployment 部署 | Capability 能力 | Tool 工具 | Multimodal 多模態 | Quantization 量化 |
|---|---|---|---|---|---|---|---|
| DeepSeek-R1 Distill | 1.5B / 7B / 8B | Transformer | Edge / GPU 邊緣 / GPU | Chain-of-thought reasoning 鏈式思考推導 | Yes 是 | Text 文字 | INT4, INT8 |
| Falcon 3 | 3B / 7B | Transformer (GQA) | Edge / GPU 邊緣端 / GPU | General NLP, reasoning 通用自然語言處理、推理 | Yes 是 | Text 文字 | INT4, INT8 |
| Falcon-E | 1B / 3B | Hybrid (Transformer + Mamba) 混合架構 (Transformer + Mamba) | MCU / Edge MCU / 邊緣端 | CPU-only inference, lowpower NLP 僅限 CPU 推論、低功耗 NLP | Ltd. 受限 | Text 文字 | INT4, INT8 |
| Gemma 2B | 2B | Transformer | Edge 邊緣 | General NLP 通用自然語言處理 | No 否 | Text 文字 | INT4, INT8 |
| Gemma 4 E2B | ~2B (eff.) | Transformer (PLE) | Edge / Mobile 邊緣 / 行動裝置 | Multimodal NLP, function calling 多模態自然語言處理 (NLP)、函式呼叫 (Function Calling) | Yes 是 | Text, Image 文字、圖片 | INT4, INT8 |
| LLaMA 3.2 | 1B / 3B | Transformer | Edge 邊緣 | General NLP, tool use 通用自然語言處理、工具使用 | Yes 是 | Text 文字 | INT4, INT8 |
| Mistral 7B | 7B | Transformer (GQA+SWA) | GPU | General NLP 通用自然語言處理 (General NLP) | Yes 是 | Text 文字 | INT4, INT8 |
| MobileLLaMA | 1.4B | Transformer (LLaMA) | Edge / Mobile 邊緣 / 行動裝置 | On-device NLP inference 裝置端自然語言處理 (NLP) 推論 | Ltd. 有限公司 | Text 文字 | INT4, INT8 |
| Phi-2 | 2.7B | Transformer | Edge 邊緣 | NLP, reasoning 自然語言處理、推理 | No 否 | Text 文字 | INT4, INT8 |
| Phi-3 Mini | 3.8B | Transformer (dense) Transformer (密集) | Edge 邊緣 (Edge) | Reasoning, coding 推理、程式碼編寫 | Yes 是 | Text 文本 | INT4, INT8, FP16 |
| Phi-3.5 Mini | 3.8B | Transformer (dense) Transformer (密集型) | Edge / Mobile 邊緣 / 行動裝置 | Multilingual NLP, long context (128K) 多語言自然語言處理 (NLP)，長文本上下文 (128K) | Yes 是 | Text 文字 | INT4, INT8, FP16 |
| Qwen2.5 | 1.5B / | Transformer | Edge 邊緣 | NLP, | Yes | Text 文字 | INT4, INT8 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| | 3B | | (Edge) | reasoning 自然語言處理 (NLP)、推理 | 是 | | |
| Qwen2.5-Coder | 1.5B / 7B | Transformer | Edge / GPU 邊緣 / GPU | Code generation, tool tasks<br>程式碼生成、工具任務 | Yes 是 | Text 文字 | INT4, INT8 |
| Qwen2.5-Omni | 3B / 7B | Transformer + Audio/Vision Enc.<br>Transformer + 音訊/視覺編碼器 | Edge / GPU 邊緣端 / GPU | Cross-modal reasoning and tool use<br>跨模態推理與工具使用 | Yes 是 | Text, Image, Audio, Video<br>文字、圖像、音訊、影片 | INT4, INT8 |
| SmolLM2 | 135M / 360M / 1.7B | Transformer | MCU / Edge MCU / 邊緣端 | Lightweight NLP; func. calling (1.7B only)<br>輕量級自然語言處理；函式呼叫（僅限 1.7B 版本） | Ltd. † 有限公司 † | Text 正文 | INT4, INT8 內攻 4，中 8 |
| SmolVLM2 | 256M / 500M | Vision-Language (VLM) 視覺語言模型 (VLM) | MCU / Edge 微控制器 (MCU) / 邊緣端 | Compact vision-language inference<br>緊湊型視覺語言推論 | Ltd. 有限公司 | Text, Image, Video 文字、圖片、影片 | INT4, INT8 |
| TinyLlama | 1.1B | Transformer (LLaMA-2) | Edge 邊緣 (Edge) | General NLP 通用自然語言處理 (General NLP) | No 否 | Text 文字 | INT4, INT8 |

† Tool use supported only in the 1.7B-Instruct variant. Models ordered alphabetically; variants by ascending parameter count.

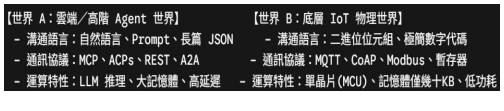
† 僅 1.7B-Instruct 版本支援工具使用（Tool use）。模型按字母順序排列；變體按參數數量升序排列。

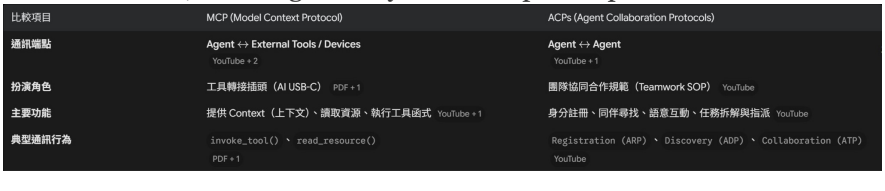
missing layer [17]. The Model Context Protocol (MCP) standardizes how an agent connects to external tools, data sources, and contextual resources through a uniform interface. The Agent-to-Agent (A2A) protocol standardizes how agents discover one another, advertise their capabilities, delegate tasks, and exchange results, and protocol suites such as ACPs extend this direction with dedicated mechanisms for agent registration, discovery, interaction, and collaboration [18]. In the reference architecture, these agent-level protocols operate between the cloud and edge tiers, while device-level protocols connect the edge and device tiers (Fig. 1).

Agent collaboration protocols

缺失層 [17]。模型上下文協定（Model Context Protocol, MCP）標準化了代理程式如何透過統一介面連接到外部工具、資料來源和上下文資源。代理程式對代理程式（Agent-to-Agent, A2A）協定標準化了代理程式如何相互發現、宣傳其能力、委派任務以及交換結果，而如 ACPs 等協定套件則透過專用的代理程式註冊、發現、互動和協作機制擴展了這一方向 [18]。在參考架構中，這些代理程式級別的協定在雲端和邊緣層之間運作，而設備級別的協定則連接邊緣和設備層（圖 1）。

Neither family can replace the other. Resource-constrained devices cannot host agent protocol stacks, and agent protocols are not built for high-frequency telemetry. The architecture therefore assigns protocol bridging to the edge tier, where edge agents translate between the two worlds in both directions. Upward, they convert raw device traffic, such as an MQTT sensor message or a CoAP response, into structured observations that agents can reason over. Downward, they turn agent decisions into concrete commands issued over whatever protocol each device understands. The bridge is more than format conversion: it aggregates, filters, and enriches information, forming the semantic boundary at which data becomes context and plans become commands. It also decouples the two sides, so the agentic layer can adopt new protocols and models without modifying the installed device base.



MCP & ACPS

這兩類協議族群無法互相取代。資源受限的設備無法承載代理程式（Agent）的協議棧，而代理程式協議也不是為了高頻遙測而設計的。因此，該架構將協議橋接分配給邊緣層，由邊緣代理程式在兩個世界之間進行雙向轉換。向上，它們將原始設備流量（例如 MQTT 感測器訊息或 CoAP 回應）轉換為代理程式可以進行推理的結構化觀測數據。向下，它們將代理程式的決策轉化為透過各設備所支援的協議發出的具體指令。此橋接不僅僅是格式轉換：它還負責聚合、過濾和豐富資訊，形成一個語義邊界，在此邊界上，數據轉化為情境，而計畫轉化為指令。它同時也解耦了雙方，使代理層能夠在不修改已安裝設備基礎的情況下，採用新的協議和模型。

協定棧就是「分工合作的網路郵務流水線」——把一包複雜的資料，一層一層包裝打包（加信封、貼地址），送達目的地後再一層一層拆開還原的完整軟體架構

E. Deployment Strategies: Device, Edge/Fog, Cloud, Hybrid

E. 部署策略：設備、邊緣/霧、雲端、混合

As shown in Fig. 1, deployment in Agentic IoT is considered through the traditional three-layer IoT architecture. A device-level deployment approach is presented at the physical/device layer, an edge/fog deployment approach at the intermediate layer, and a cloud deployment approach at the intelligence layer. The layer in which an agent will be implemented and executed can be determined by considering the requirements of the application domain, such as decision scope, response time, and the amount of data to be transferred. Cloud deployment offers the most comprehensive reasoning capacity due to the use of large models, long-term memory, and system-wide visibility. This deployment is suitable for strategic and latency-tolerant tasks such as optimization, policy generation, and cross-site analytics.

如圖 1 所示，代理式物聯網（Agentic IoT）的部署是透過傳統的三層物聯網架構來考慮的。設備級部署方法呈現於物理/設備層，邊緣/霧部署方法位於中間層，而雲端部署方法則位於智慧層。代理程式將在哪一層實作與執行，可藉由考量應用領域的需求來決定，例如決策範圍、回應時間以及待傳輸的數據量。雲端部署由於使用了大型模型、長期記憶和全系統的可視性，提供了最全面的推理能力。這種部署適用於策略性且容忍延遲的任務，例如優化、策略生成和跨站點分析。

In edge/fog deployment, priorities change. Quantized medium-sized language models (Table II) running on gateways and edge servers can respond within local latency limits, keep data local, and continue operating when the uplink is interrupted. However, this approach has to operate with more limited model capacity, memory, and context length. Therefore, edge placement becomes more suitable for locally scoped, repetitive, time-critical, or privacy-sensitive decisions rather than open-ended and system-wide reasoning. In on-

在邊緣/霧部署中，優先順序會發生變化。運行在閘道器和邊緣伺服器上的量化中型語言模型（表 II）可以在本地延遲限制內做出回應，保持數據在地化，並在向上鏈路中斷時繼續運作。然而，這種方法必須在較受限的模型容量、記憶體和上下文長度下運作。因此，邊緣部署更適合局部範圍、重複性、時間關鍵或隱私敏感的決策，而非開放式且全系統的推理。在設備上——

device deployment, on-premises solutions can be provided by considering data privacy and security, as well as fast response time, mostly for small language models.

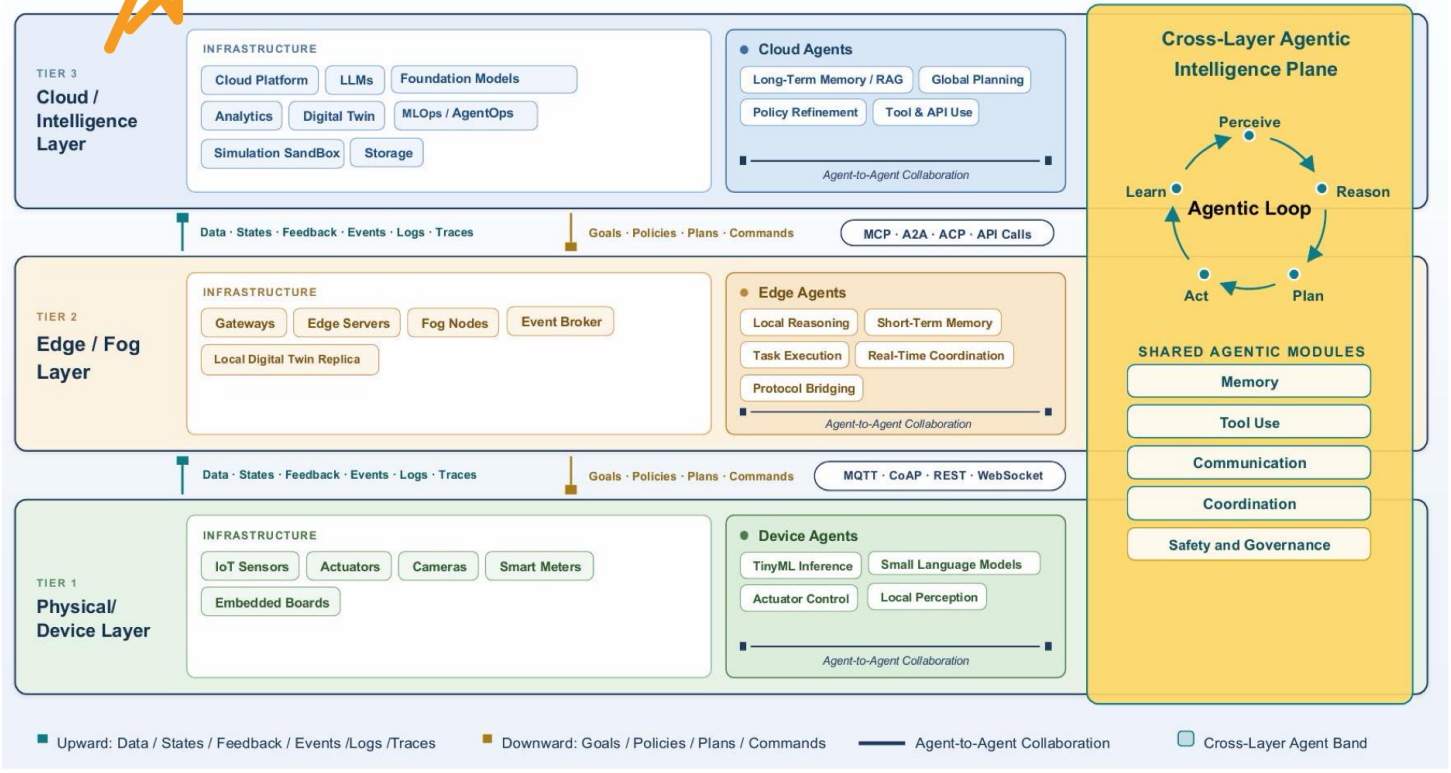
在設備部署方面，考量到數據隱私與安全性，以及快速的反應時間，可以提供地端解決方案，這類方案多半針對小型語言模型。

On the other hand, instead of using each layer alone for deployment, we present an architecture that combines them in a hybrid manner. Here, we position hybrid deployment not as a simple compromise between the cloud and the device, but as the fundamental operating model for Agentic IoT systems. In this structure, responsibilities are distributed across layers. In this deployment, device agents perform immediate perceptionaction behaviors, edge agents make tactical decisions in line with the policies received from upper layers, and cloud agents carry out strategic planning and long-term learning processes. The transfer of tasks to upper layers is performed according to the requirements of the relevant task. Therefore, while routine situations can be resolved locally, novel or high-risk situations are transferred to upper layers because they require more comprehensive reasoning. In this way, more comprehensive reasoning is performed by tolerating higher latency. In cases where connectivity is interrupted, edge and device agents continue to operate according to the most recently delivered policies, and the system experiences gradual performance degradation instead of stopping completely. This hierarchical distribution of intelligence enables Agentic IoT systems to respond quickly on the device and edge side, while also benefiting from deeper cloud-scale reasoning capacity.

另一方面，我們提出了一種以混合方式結合各層級的架構，而非單獨使用每一層進行部署。在此，我們將混合部署定位為代理式物聯網（Agentic IoT）系統的基礎運作模式，而不僅僅是雲端與設備之間的簡單折衷。在這種結構中，職責分布於各個層級。在此部署模式下，設備代理執行即時的感知與行動行為；邊緣代理根據從上層接收到的策略做出戰術決策；而雲端代理則負責執行策略規劃與長期學習過程。任務向上層的轉移是根據相關任務的需求來進行的。因此，雖然常規情況可以在本地解決，但新穎或高風險的情況則會轉移到上層，因為它們需要更全面的推理。透過這種方式，系統在容忍較高延遲的情況下執行更全面的推理。在連線中斷的情況下，邊緣和設備代理會根據最近交付的策略繼續運作，系統會經歷逐漸的性能降級，而非完全停止。這種智慧的階層式分布使代理式物聯網系統能夠在設備和邊緣端快速反應，同時也能受益於更深層的雲端級推理能力。

根據推理的需求作分配算力：常規本地

為了效能!!


TIER 3
Cloud / Intelligence Layer
INFRASTRUCTURE
Cloud Platform
LLMs
Foundation Models
Analytics
Digital Twin
MLOps / AgentOps
Simulation SandBox
Storage
Cloud Agents
Long-Term Memory / RAG
Global Planning
Policy Refinement
Tool & API Use
Agent-to-Agent Collaboration
Data · States · Feedback · Events · Logs · Traces
Goals · Policies · Plans · Commands
MCP · A2A · ACP · API Calls
TIER 2
Edge / Fog Layer
INFRASTRUCTURE
Gateways
Edge Servers
Fog Nodes
Event Broker
Local Digital Twin Replica
Edge Agents
Local Reasoning
Short-Term Memory
Task Execution
Real-Time Coordination
Protocol Bridging
Agent-to-Agent Collaboration
Data · States · Feedback · Events · Logs · Traces
Goals · Policies · Plans · Commands
MQTT · CoAP · REST · WebSocket
TIER 1
Physical/ Device Layer
INFRASTRUCTURE
IoT Sensors
Actuators
Cameras
Smart Meters
Embedded Boards
Device Agents
TinyML Inference
Small Language Models
Actuator Control
Local Perception
Agent-to-Agent Collaboration
Cross-Layer Agentic Intelligence Plane
Perceive
Reason
Plan
Act
Learn
Agentic Loop
SHARED AGENTIC MODULES
Memory
Tool Use
Communication
Coordination
Safety and Governance
Upward: Data / States / Feedback / Events /Logs /Traces
Downward: Goals / Policies / Plans / Commands
Agent-to-Agent Collaboration
Cross-Layer Agent Band

## IV. Domain-Specific Agentic IoT Applications

## IV. 特定領域的代理式物聯網應用

This section reviews existing agentic IoT systems and approaches proposed in recent studies. These works focus on enabling autonomous behavior, coordination, and intelligent decision making through various frameworks and models. The approaches are analyzed in terms of their methodology, architecture and key results.

本節回顧了近期研究中提出的現有代理式物聯網系統與方法。這些研究重點在於透過各種框架與模型，實現自主行為、協調與智慧決策。本文從方法論、架構及關鍵結果等方面對這些方法進行了分析。

### A. IoT & Agent Infrastructure

### A. 物聯網與代理程式基礎設施

In this subsection, we examine studies focusing on potential agent architectures and deployment infrastructures within the scope of agentic IoT.

在本小節中，我們將探討在代理型物聯網（Agentic IoT）範疇內，聚焦於潛在代理程式架構與部署基礎設施的相關研究。

Yu et al. [19] propose an Internet of Agents paradigm to address the limited adoption of multi-agent systems in IoT. They highlight the complexity of existing approaches and emphasize the need for accessible tools for end-users. They review existing agent-oriented models and tools such as JADE, the BDI model, Prometheus, Tropos, and Goal Net, showing that they are mainly designed for expert developers.The approach is based on intelligent agents, emphasizing reasoning, autonomy, and user-driven customization. The study concludes that simplifying these models and enabling end-user participation is essential for wider adoption of agent-based IoT systems. PicoValencia et al. [20] propose a semantic agent contract model for IoA to improve interoperability among agents. They use an OWL-based ontology (IoA-OWL) with linked data to describe agents, services, and context. The model defines standardized semantic profiles, including agent, context, service, social, model, and object layers, to represent agent capabilities and relationships. The approach is based on semantic (ontologydriven) agents, enabling structured knowledge representation and consistent interaction across heterogeneous systems. The study concludes that this model enhances interoperability and coordination in IoA environments.Pico-Valencia et al. [21] propose a MAS-ROA architecture to integrate multi-agent systems with IoT for managing smart objects. They use RESTful resource-oriented services and workflow-driven agents to enable adaptive and collaborative control. The architecture follows a multi-layer design where agents interact with IoT resources through RESTful services and dynamic workflows. The approach is based on reactive agents within a multiagent system, where agent behavior is driven by workflowbased control and stimulus-response mechanisms. The study shows that this model improves adaptability, interoperability, and collaborative control in IoT environments. Pico-Valencia et al. [22] analyze the IoT paradigm from an intelligence and autonomy perspective, proposing the Internet of Agents as an evolution to enhance these capabilities. They examine multiple application domains and present IoA-based adaptations of existing IoT systems. The study analyzes 24 IoT application scenarios and redesigns them using agent-based approaches across domains such as smart industry, smart city, and healthcare. The approach is based on intelligent software agents. It focuses on autonomy, learning, and collaboration to improve system behavior. The study concludes that adding these agents to IoT environments can significantly improve intelligence, adaptability, and interoperability. Pico-Valencia et al. [23] propose a systematic method for building IoA systems from existing IoT infrastructures. They use a bottom-up and agile-based approach. In this approach, IoT objects are modeled as agents called LOAs. The method also uses semantic contracts and workflows. It is organized into iterative phases. These phases help transform IoT components into agent-based systems and support continuous improvement. The approach is based on semantic, ontology-driven agents. LOAs interact by using structured knowledge and workflow coordination. The study shows that this method improves scalability, adaptability, and interoperability in IoA system development. RodriguezBenitez et al. [24] propose an IoA-based architecture for distributed training and deployment of CNN models in IoT environments. They use a FIPA-compliant multi-agent system with specialized agents (e.g., Trainer, Deployment, Monitor) and communication via MQTT and FIPA-ACL. The system applies model quantization and TCP-based transmission to enable execution on resource-constrained devices, achieving efficient

1. 什麼是 IoA？（一句話降維）IoA 全名是 Internet of Agents，中文翻作「代理人聯網」或「智能體互聯網」。 傳統物聯網（IoT，Internet of Things）：連上網路的是「硬體設備」（感測器、開關、攝影機、晶圓機台儀表）。 代理人聯網（IoA，Internet of Agents）：連上網路的是一個個「具有思考、規劃與行動能力的 AI Agent（大腦）」。 簡單來說：IoT 是「設備連網」，IoA 則是讓全世界不同功能的「AI 專家跨系統組隊連網」。
2. 生產線與研究室白話比喻想像一下半導體廠或是我們做智慧系統的大專案：傳統做法（單一系統或單隻 Agent）：你叫同一個助理（單隻 Agent）去既看晶圓厚度、又寫排程演算法、還要去查庫存和回客服，它一定會顧此失彼、上下文塞爆當機。
IoA（代理人聯網的世界）：
工廠裡不需要一個全能超人，而是建立一個「AI 專家公會（IoA 網路）」： 廠內有專精影像瑕疵檢測的「視覺 Agent」。有精通機台調校參數的「製程 Agent」。甚至能跨網路呼叫外部供應商的「零件採購 Agent」。當主管（使用者）下達一個高階目標：「今天 3 號機出現微刮痕，請在下午前排除並補足耗材」，這群分散在不同伺服器上的 Agent 會透過網路互相打招呼、找幫手、分工認領任務，最後把整件事辦妥回報！這整套讓各路 AI 代理人像網頁伺服器一樣互聯互通的架構，就叫做 IoA。

3. IoA 的三大關鍵運作機制在論文中，IoA 通常圍繞以下三個核心運作： 同伴尋找（Discovery）：Agent 上線後，能去通訊錄查找「現在網路上誰懂化學機械研磨（CMP）？」或「誰有權限開關這台冷氣？」 協商與派工（Task Decomposition & Delegation）：把大任務拆成子任務，發給最擅長的 Agent 平行處理。 標準通訊語言（Protocols）：大家不再自說自話，而是用像 ACPs 或 A2A 這類標準協議傳遞訊息，確保各家開發的 Agent 溝通不會雞同鴨講。

and accurate deployment.

Yu 等人 [19] 提出了「代理人互聯網」（Internet of Agents）範式，旨在解決多代理人系統在物聯網（IoT）中採用率有限的問題。他們強調了現有方法的複雜性，並指出終端用戶需要更易於使用的工具。他們回顧了現有的導向代理人模型與工具，如 JADE、BDI 模型、Prometheus、Tropos 和 Goal Net，並指出這些工具主要是為專業開發者設計的。該方法基於智慧代理人，強調推理、自主性與用戶驅動的客製化。研究結論指出，簡化這些模型並讓終端用戶參與，對於推廣基於代理人的物聯網系統至關重要。 Pico-Valencia 等人 [20] 為 IoA 提出了一種語義代理人合約模型，以提高代理人之間的互操作性。他們使用基於 OWL 的本體（IoA-OWL）結合連結數據來描述代理人、服務與情境。該模型定義了標準化的語義剖面，包括代理人層、情境層、服務層、社交層、模型層和物件層，用以表示代理人的能力與關係。此方法基於語義（本體驅動）代理人，實現了結構化的知識表示以及跨異質系統的一致性互動。研究結論顯示，該模型增強了 IoA 環境中的互操作性與協調性。 Pico-Valencia 等人 [21] 提出了一種 MAS-ROA 架構，將多代理人系統與物聯網整合，以管理智慧物件。他們利用 RESTful 資源導向服務與工作流驅動的代理人來實現自適應與協作控制。該架構採用多層設計，代理人透過 RESTful 服務與動態工作流與物聯網資源進行互動。此方法基於多代理人系統中的反應式代理人，其行為由基於工作流的控制與刺激-反應機制驅動。研究顯示，該模型提高了物聯網環境中的自適應性、互操作性與協作控制能力。 Pico-Valencia 等人 [22] 從智慧化與自主性的角度分析了物聯網範式，並提出將代理人互聯網作為提升這些能力的演進方向。他們檢視了多個應用領域，並展示了現有物聯網系統基於 IoA 的改編方案。該研究分析了 24 個物聯網應用場景，並使用基於代理人的方法在智慧工業、智慧城市和醫療保健等領域進行重新設計。此方法基於智慧軟體代理人，專注於自主性、學習與協作，以改善系統行為。研究結論指出，在物聯網環境中加入這些代理人可以顯著提升智慧化、自適應性與互操作性。 Pico-Valencia 等人 [23] 提出了一種從現有物聯網基礎設施構建 IoA 系統的系統化方法。他們採用了由下而上且基於敏捷開發的方法。在這種方法中，物聯網物件被建模為稱為 LOA 的代理人。該方法還使用了語義合約與工作流，並以迭代階段的方式組織。 這些階段有助於將 IoT 組件轉化為基於代理的系統，並支持持續改進。該方法基於語義、本體驅動的代理。LOA 透過使用結構化知識和工作流協調進行互動。研究顯示，此方法提高了 IoA 系統開發中的可擴展性、適應性和互操作性。RodriguezBenitez 等人 [24] 提出了一種基於 IoA 的架構，用於在 IoT 環境中進行 CNN 模型的分布式訓練和部署。他們使用符合 FIPA 標準的多代理系統，包含專門的代理（例如：訓練代理、部署代理、監控代理），並透過 MQTT 和 FIPA-ACL 進行通訊。該系統應用模型量化和基於 TCP 的傳輸，使其能在資源受限的設備上執行，實現了高效且準確的部署。

B. Security & Trust

B. 安全與信任

In this subsection, we examine studies focusing on threat detection, trust establishment, and secure coordination.

在本小節中，我們檢視專注於威脅檢測、信任建立和安全協調的研究。

Aref et al. [25] propose a trust establishment model (ATeIF) for IoA that enables agents to adapt their behavior using implicit feedback. They use retention-based metrics and reinforcement learning concepts to estimate trust without explicit user input. The study models trust using multi-criteria

Aref 等人 [25] 為 IoA 提出了一種信任建立模型 (ATeIF)，使代理能夠利用隱性回饋來調整其行為。他們使用基於留存的指標和強化學習概念，在沒有使用者明確輸入的情況下估算信任度。該研究使用多標準來建模信任。

service evaluation and retention-based learning to adjust agent behavior dynamically. The approach is based on intelligent agents within a multi-agent system (MAS). The study concludes that the proposed model improves trustworthiness and increases successful interactions among agents. Aref et al. [26] propose an integrated trust establishment model for IoA that combines multiple feedback mechanisms to improve agent behavior. They use both direct and indirect trust signals. These signals help adjust agent interactions dynamically. The study combines reputation, experience, and contextual factors to calculate trust values in a more complete way. The approach is based on intelligent agents in a multi-agent system. The study concludes that the proposed model improves trust accuracy and supports more reliable interactions among agents. Li et al. [18] propose a unified protocol framework called ACPs for the Internet of Agents. The aim is to solve interoperability and coordination problems. They define standard protocols for agent registration, agent discovery, agent interaction, and agent collaboration. These protocols include ARP, ADP, AIP, and ATP. The study introduces a layered IoA architecture and supports capability orchestration and task decomposition across heterogeneous agents. It also considers the integration of large language model (LLM)-based agents to enhance communication and intelligent interaction. The approach is based on intelligent, heterogeneous multi-agent systems (MAS), enabling scalable and coordinated agent interactions. The study concludes that the proposed framework improves interoperability, collaboration, and adaptability in IoA environments. Kumia et al. [27] propose a dual-proxy, multi-plane gateway architecture to secure agentic AI in IoT systems. They use separate planes for data, control, security, and coordination, along with a sentinel mechanism and digital twin for safe execution. The study introduces a seven-plane architecture and a goal-plan-step validation mechanism to ensure safe and verifiable agent actions. It incorporates LLM-based agents and supports interaction with models such as GPT-4, along with tool-augmented frameworks and RAG-based context integration. The approach is based on intelligent, agentic AI systems integrated with IoT environments. The study concludes that the proposed architecture improves security, governance, and reliability in agentic IoT systems. Ren et al. [28] propose the Internet of Agentic AI (IoA) paradigm to enable scalable and intent-driven coordination among agents. They introduce a layered architecture with core protocols and an intelligencebased reinforcement learning (IRL) framework for multi-agent coordination. The study enables dynamic task composition, semantic interaction, and distributed planning among heterogeneous agents. It also incorporates cognitive capabilities supported by LLM-based agents and tool-augmented frameworks (e.g., AutoGen, CrewAI, LangGraph). The approach is based on intelligent, heterogeneous multi-agent systems (MAS) combined with

reinforcement learning. The study concludes that the proposed IoA paradigm improves adaptability, coordination efficiency, and scalability. Vijetha [29] proposes a selfadaptive agentic framework called AISAF. AISAF is designed for threat detection in cloud, edge, and IoT systems. The approach uses a hybrid CNN-LSTM Transformer model. It also uses a drift-aware meta-optimization mechanism for selective adaptation. The framework detects changes in the latent space and monitors confidence scores. These mechanisms allow the system to trigger model updates automatically. They also help the system maintain performance when data distributions change over time. The framework includes an attention-based explainability layer. This layer makes the decision-making process more transparent and consistent during adaptation. The approach is based on intelligent agentic AI systems in a heterogeneous multi-agent environment. The study shows that AISAF improves adaptability, robustness, and recovery from data drift in dynamic cybersecurity environments.

服務評估與基於保留的學習，以動態調整代理人行為。該方法基於多代理人系統（MAS）中的智慧代理人。研究結論指出，所提出的模型提升了可信度，並增加了代理人之間成功互動的次數。Aref 等人 [26] 為 IoA 提出了一種整合式信任建立模型，結合多種回饋機制以改善代理人行為。他們同時使用直接與間接的信任信號，這些信號有助於動態調整代理人的互動。該研究結合了聲譽、經驗與情境因素，以更完整的方式計算信任值。此方法基於多代理人系統中的智慧代理人。研究結論顯示，該模型提高了信任準確性，並支持代理人之間更可靠的互動。Li 等人 [18] 為代理人網路（Internet of Agents）提出了一個名為 ACPs 的統一協定框架，旨在解決互操作性與協調問題。他們定義了代理人註冊、代理人發現、代理人互動及代理人協作的標準協定，包括 ARP、ADP、AIP 與 ATP。該研究介紹了一種分層的 IoA 架構，並支持跨異質代理人的能力編排與任務分解。此外，它還考慮了整合基於大型語言模型（LLM）的代理人，以增強通訊與智慧互動。此方法基於智慧異質多代理人系統（MAS），實現了具擴展性且協調的代理人互動。研究結論指出，該框架提升了 IoA 環境中的互操作性、協作性與適應性。Kumia 等人 [27] 提出了一種雙代理、多平面的閘道器架構，以確保物聯網系統中代理式 AI 的安全。他們為數據、控制、安全與協調使用獨立的平面，並結合哨兵機制與數位分身以確保安全執行。該研究引入了七平面架構以及「目標-計畫-步驟」驗證機制，以確保代理人行為安全且可驗證。它納入了基於 LLM 的代理人，並支持與 GPT-4 等模型的互動，同時結合了工具增強框架與基於 RAG 的情境整合。此方法基於與物聯網環境整合的智慧代理式 AI 系統。研究結論顯示，該架構提升了代理式物聯網系統的安全性、治理能力與可靠性。Ren 等人 [28] 提出了代理式 AI 網路（IoA）範式，以實現代理人之間具擴展性且意圖驅動的協調。他們引入了帶有核心協定的分層架構，以及用於多代理人協調的智慧強化學習（IRL）框架。該研究實現了異質代理人之間的動態任務組合、語義互動與分散式規劃。它還納入了由基於 LLM 的代理人與工具增強框架（例如 AutoGen、CrewAI、LangGraph）所支持的認知能力。 此方法基於結合強化學習的智慧型異質多代理人系統（MAS）。研究結論指出，所提出的 IoA 範式提升了適應性、協調效率與擴展性。Vijetha [29] 提出了一種名為 AISAF 的自適應代理框架。AISAF 專為雲端、邊緣及 IoT 系統中的威脅檢測而設計。該方法採用混合 CNN-LSTM Transformer 模型，並使用具備漂移感知能力的元優化機制來進行選擇性適應。該框架能檢測潛在空間的變化並監控置信度分數。這些機制使系統能夠自動觸發模型更新，並在數據分佈隨時間變化時協助系統維持效能。此外，該框架包含一個基於注意力機制的解釋性層，使適應過程中的決策過程更加透明且一致。此方法基於異質多代理人環境中的智慧代理 AI 系統。研究顯示，AISAF 在動態網路安全環境中提升了適應性、穩健性以及從數據漂移中恢復的能力。

- **ARP**（代理人註冊協定）
- **ADP**（代理人發現協定）
- **AIP**（代理人互動協定）
- **ATP**（代理人協作協定）

C. Network C. 網路

In this subsection, we examine network-based studies focusing on distributed and edge-computing-based learning, as well as intelligent resource and task management.

在本小節中，我們將檢視聚焦於分散式與邊緣運算學習，以及智慧資源與任務管理的網路相關研究。

Pico-Valencia et al. [31] analyze how IoT systems are evolving toward the Internet of Agents paradigm. The goal is to make IoT systems more autonomous and intelligent. They examine the limitations of traditional IoT architectures. They also propose agent-based adaptations to support more flexible and intelligent system behavior. The study presents a layered IoA architecture. It also identifies the main components needed for agent integration. These components include communication, coordination, and semantic interaction mechanisms. The study explains how agent-based transformation can improve system responsiveness and decision-making. The approach is based on intelligent agents in a multi-agent system. The study concludes that IoA can significantly improve adaptability, interoperability, and autonomy in IoT environments. Jeong et al. [32] propose DRACO, a decentralized and asynchronous federated learning framework for distributed AI in IoT environments. They design an agent-based system that allows devices to train models together without central coordination. The system supports asynchronous updates and peer to peer communication. The framework includes mechanisms for dynamic participation, model aggregation, and communication efficiency under heterogeneous conditions. It also improves scalability and robustness by reducing dependence on central servers and handling device differences. The approach is based on intelligent agents in a decentralized multi-agent setting. The study concludes that DRACO improves scalability, efficiency, and resilience in distributed IoT learning systems. Jiang et al. [33] propose a Stackelberg game-based framework for agentic AI task offloading in the Internet of Agents. The goal is to address resource limitations. They model interactions among wireless, mobile, fixed, and aerial agents. This allows hierarchical and efficient task delegation. The study formulates the problem as a leader-follower game to optimize pricing and offloading strategies under heterogeneous conditions. It considers agentic AI tasks supported by large models such as LLMs and VLMs for reasoning and decision-making. The approach is based on intelligent, heterogeneous multiagent systems. The study

concludes that the method improves

Pico-Valencia 等人 [31] 分析了物聯網系統如何朝向代理人聯網（Internet of Agents, IoA）範式演進。其目標是使物聯網系統更具自主性與智慧化。他們檢視了傳統物聯網架構的局限性，並提出基於代理人的適應方案，以支持更靈活且智慧的系統行為。該研究提出了一種分層的 IoA 架構，並識別了代理人整合所需的主要組件，包括通訊、協調與語義交互機制。研究說明了基於代理人的轉型如何提升系統的響應能力與決策水準。此方法基於多代理人系統中的智慧代理人。研究結論指出，IoA 能顯著提升物聯網環境中的適應性、互操作性與自主性。Jeong 等人 [32] 提出了 DRACO，這是一個用於物聯網環境中分散式人工智慧的去中心化且非同步的聯邦學習框架。他們設計了一個基於代理人的系統，允許設備在沒有中央協調的情況下共同訓練模型。該系統支持非同步更新與點對點通訊。該框架包含動態參與、模型聚合以及在異質條件下提升通訊效率的機制。它還透過減少對中央伺服器的依賴並處理設備差異，提升了可擴展性與穩健性。此方法基於去中心化多代理人環境中的智慧代理人。研究結論指出，DRACO 提升了分散式物聯網學習系統的可擴展性、效率與韌性。Jiang 等人 [33] 針對代理人聯網中的代理式 AI 任務卸載，提出了一個基於史塔克伯格賽局（Stackelberg game）的框架。其目標是解決資源限制問題。他們對無線、行動、固定與空中代理人之間的交互進行建模，從而實現階層化且高效的任務委派。該研究將問題公式化為領導者-追隨者賽局，以優化異質條件下的定價與卸載策略。它考慮了由 LLMs 和 VLMs 等大型模型支持的代理式 AI 任務，用於推理與決策。此方法基於智慧且異質的多代理人系統。研究結論指出，該方法改善了

Table 3: TABLE III 表 3：表 III
Comparative Analysis of Internet of Agents and Agentic IoT Studies Across IoT Application Domains

代理人聯網與代理式物聯網研究在各物聯網應用領域的比較分析

| IoT Domain 物聯網領域 | Study 研究 | Year 年份 | Problem Context 問題背景 | Taxonomy 分類法 | Architecture 架構 | Underlying Mechanism 底層機制 |
|---|---|---|---|---|---|---|
| Infrastructure 基礎設施 | Yu et al. [19]<br>Yu 等人 [19] | 2013 | Resource Mgmt. 資源管理 | Cognitive 認知 | Intelligent 智慧 | BDI (Belief-Desire-Intention)<br>BDI (信念-欲望-意圖) |
| | Pico-Valencia and HolgadoTerriza [20]<br>Pico-Valencia 與 HolgadoTerriza [20] | 2016 | Interoperability 互通性 | Cognitive 認知 | Semantic 語義 | BDI (Belief-Desire-Intention)<br>BDI (信念-欲望-意圖) |
| | Pico-Valencia and HolgadoTerriza [30]<br>Pico-Valencia 與 HolgadoTerriza [30] | 2018 | System Integration 系統整合 | Cognitive 認知 | Reactive 反應式 | N/A 不適用 |
| | Pico-Valencia et al. [22]<br>Pico-Valencia 等人 [22] | 2018 | Architecture Analysis 架構分析 | Operational 運作層面 | Software 軟體 | FIPA-ACL (Agent Comm. Lang.)<br>FIPA-ACL (代理人通訊語言) |
| | Pico-Valencia et al. [23]<br>Pico-Valencia 等人 [23] | 2019 | Interoperability 互通性 | Cognitive 認知 | Semantic 語義 | LOA (Linked Open Agent)<br>LOA (連結開放代理) |
| | Rodriguez-Benitez et al. [24]<br>Rodriguez-Benitez 等人 [24] | 2022 | Deep Learning Ops 深度學習營運 (Deep Learning Ops) | Operational 營運層面 | Specialized 專業化 | TFLite (Tensorflow Lite) |
| Security 安全性 | Aref and Tran [25]<br>Aref 與 Tran [25] | 2017 | Trust Modeling 信任建模 | Distributed 分散式 | Heterogeneous 異質性 | Implicit Feedback 隱性回饋 |
| | Aref and Tran [26]<br>Aref 與 Tran [26] | 2020 | Trust Establishment 信任建立 | Cognitive 認知 | Intelligent 智慧 | Q-Learning |
| | Li et al. [18]<br>Li 等人 [18] | 2025 | Multi-Agent Collab. 多代理人協作 | Distributed 分散式 | Heterogeneous 異質性 | Protocol-Based 基於協定 |
| | Kumi et al. [27]<br>Kumi 等人 [27] | 2025 | Governance 治理 | Autonomous 自主 | Agentic AI 代理式 AI (Agentic AI) | LLM-based Reasoning 基於 LLM 的推理 |
| | Ren et al. [28] | 2026 | Protocol Design 協定設計 | Protocol-Driven 協定 | Layered IoA 分 | RL (Reinforcement |

| | | | | | | |
|---|---|---|---|---|---|---|
| | Ren 等人 [28] | | Design 協定設計 | Driven 協定驅動 | 層 IoA | (Reinforcement Learning) RL（強化學習） |
| | Vijetha [29] | 2026 | Threat Detection 威脅偵測 | Autonomous 自主式 | Agentic AI 代理式 AI | CNN-LSTM-Transformer |
| Network 網路 | Pico-Valencia et al. [31] Pico-Valencia 等人 [31] | 2019 | Data Interoperability 資料互通性 | Cognitive 認知 | Intelligent 智慧型 | LOA (Linked Open Agent) |
| | Jeong and Kountouris [32] Jeong 與 Kountouris [32] | 2025 | Federated Learning 聯邦學習 | Distributed 分散式 | Decentralized 去中心化 | SGD (Stoch. Grad. Descent) SGD (隨機梯度下降) |
| | Jiang et al. [33] Jiang 等人 [33] | 2025 | Task Offloading 任務卸載 | Operational 運作面 | Embodied AI 具身智慧 | GT (Game-Theoretic Model) GT（賽局理論模型） |
| | Elewah et al. [34] Elewah 等人 [34] | 2025 | Real-time Search 即時搜尋 | Distributed 分散式 | MAS | RAG (Retrieval-Augm. Gen.) RAG (檢索增強生成) |
| | Vashisht et al. [35] Vashisht 等人 [35] | 2026 | Network Slicing 網路切片 | Distributed 分散式 | Supervisory 監控式 | Ensemble Learning 整合學習 |
| Smart Cities 智慧城市 | Bui and Jung [36] Bui 與 Jung [36] | 2018 | Traffic Control 交通控制 | Cognitive 認知 | Intelligent 智慧 | GME (Group Mutual Exclusion) GME (群組互斥) |
| | Kamel [37] | 2025 | Disaster Response 災害應變 | Distributed 分散式 | Heterogeneous 異質性 | YOLOv8 (Object Detection) YOLOv8 (物件偵測) |
| | Chumyen [38] | 2025 | Water Distribution 配水系統 | Distributed 分散式 | Decentralized 去中心化 | PPO (Prox. Policy Optim.) PPO (近端策略優化) |
| Industry 工業 | Tariq et al. [39] Tariq 等人 [39] | 2025 | Precision Farming 精準農業 | Cognitive 認知 | Rule-based 基於規則 | MiT-B0 (Vision Transformer) |
| | Kandamali et al. [40] Kandamali 等人 [40] | 2025 | Irrigation Advisory 灌溉建議 | Autonomous 自主式 | Tool-Use LLM 工具使用 LLM | LLM (Llama 3.1) |
| | Sümer et al. [41] Sümer 等人 [41] | 2025 | Machine Customers 機器客戶 | Semi-autonomous 半自主 | Blockchain-IoT 區塊鏈物聯網 | RBS (Rule-Based System) RBS (規則導向系統) |
| | Petrović et al. [42] Petrović 等人 [42] | 2025 | Sustainability 永續性 | Design-time 設計階段 | Autonomous 自主式 | LLM (GPT-4o) |

| | | | | | | |
|---|---|---|---|---|---|---|
| | Mohanaprasad et al. [43]<br>Mohanaprasad 等人 [43] | 2026 | Smart Home Orch. 智慧家庭編排 | Autonomous 自主性 | Multimodal 多模態 | VLM (Qwen 2.5 VL) |

resource efficiency and task execution performance. Elewah et al. [34] propose an agentic search engine framework for real-time information retrieval in dynamic environments. They design an agent-based system that enables autonomous query understanding, task decomposition, and multi-source information aggregation. The framework supports real-time decisionmaking and adaptive search through coordinated agent interactions and context-aware processing. It incorporates LLMbased components to enhance natural language understanding and reasoning during query execution. The approach is based on intelligent and heterogeneous multi agent systems. The study concludes that the proposed system improves search accuracy, responsiveness, and adaptability in real time scenarios. Vashisht et al. [35] propose an agentic intelligenceassisted machine learning fusion framework for sustainable 6G network slicing. The framework is designed for dynamic and multimodal conditions. They use a stacked ensemble model that combines LightGBM, Random Forest, and Logistic Regression. They also use an agentic overlay to enforce SLA constraints in real time. The framework combines multimodal features and supports adaptive decision-making through a hybrid machine learning and agentic architecture. It allows slicing decisions to be checked and adjusted according to SLA requirements. These requirements include throughput, latency, and energy consumption. The approach is based on intelligent agents in a multi agent system. The study concludes that the proposed framework improves accuracy, stability, and SLA compliance in 6G environments.

資源效率與任務執行效能。Elewah 等人 [34] 提出了一種代理式搜尋引擎框架，用於動態環境中的即時資訊檢索。他們設計了一個基於代理的系統，能夠實現自主查詢理解、任務分解以及多源資訊聚合。該框架透過協調的代理互動與上下文感知處理，支援即時決策與自適應搜尋。它整合了基於大型語言模型（LLM）的組件，以增強查詢執行過程中的自然語言理解與推理能力。此方法基於智慧且異質的多代理系統。研究結論指出，該系統提高了即時場景下的搜尋準確性、響應速度與適應性。Vashisht 等人 [35] 提出了一種代理式智慧輔助機器學習融合框架，用於永續的 6G 網路切片。該框架專為動態與多模態條件設計。他們使用了一種結合 LightGBM、隨機森林（Random Forest）與邏輯迴歸（Logistic Regression）的堆疊整合模型（Stacked Ensemble Model）。他們還利用代理式疊加層（Agentic Overlay）來即時執行服務水準協議（SLA）限制。該框架結合了多模態特徵，並透過混合機器學習與代理式架構支援自適應決策。它允許根據 SLA 要求檢查並調整切片決策。這些要求包括吞吐量、延遲與能耗。此方法基於多代理系統中的智慧代理。研究結論指出，所提出的框架提高了 6G 環境下的準確性、穩定性與 SLA 合規性。

D. Smart Cities & Disaster

D. 智慧城市與災害

In this subsection, we present studies that address issues such as emergency response, disaster management, and mobility in smart cities within the IoA framework.

在本小節中，我們介紹在 IoA 框架下解決智慧城市中緊急應變、災害管理及移動性等問題的研究。

Bui et al. [36] propose an Internet of Agents framework for connected vehicles. The goal is to support distributed traffic control without using centralized infrastructure. In this framework, each vehicle is modeled as an agent. Vehicles communicate with each other through vehicle to vehicle communication. They coordinate their decisions by using an extended Ricart Agrawala algorithm based on group mutual exclusion. The approach is based on intelligent agents in a multi-agent system. The study concludes that the proposed framework improves traffic efficiency and reduces waiting time compared to traditional traffic systems. Kamel et al. [37] propose an integrated IoA - IoT framework called AIMS-AI

Bui 等人 [36] 為聯網車輛提出了一種代理人網路（Internet of Agents）框架。其目標是在不使用集中式基礎設施的情況下支援分散式交通控制。在此框架中，每輛車都被建模為一個代理人。車輛透過車對車（V2V）通訊進行交流。它們利用基於群組互斥的擴展型 Ricart Agrawala 演算法來協調決策。該方法基於多代理人系統中的智慧代理人。研究結論指出，與傳統交通系統相比，所提出的框架提高了交通效率並減少了等待時間。Kamel 等人 [37] 提出了一種名為 AIMS-AI 的 IoA - IoT 整合框架，

for smart disaster detection and response. They design a multi layered architecture for real time sensing and coordinated decision making among heterogeneous agents. The framework uses AI models such as YOLOv8, BERT, and large language models to improve perception and response. The approach is based on intelligent and heterogeneous multi-agent systems. The study concludes that AIMS-AI improves scalability and real-time response performance. Chumyen et al. [38] propose a multi-agent IoT-based smart water distribution system for disaster relief in smart cities. They design autonomous dispenser agents integrating computer vision (MobileNet-SSD) and LSTM-based demand forecasting to enable real-time perception and prediction. The system uses multi-agent reinforcement learning (PPO) for decentralized coordination under resource-constrained environments. The approach is based on intelligent, heterogeneous multi-agent systems. The study concludes

that it improves fairness, efficiency, and resilience in emergency water distribution.

用於智慧災害偵測與應變。他們設計了一種多層架構，用於異質代理人之間的即時感測與協調決策。該框架使用 YOLOv8、BERT 和大型語言模型等 AI 模型來提升感知與應變能力。該方法基於智慧且異質的多代理人系統。研究結論指出，AIMS-AI 提升了擴展性與即時應變效能。Chumyen 等人 [38] 提出了一種基於多代理人 IoT 的智慧配水系統，用於智慧城市的災難救援。他們設計了整合電腦視覺（MobileNet-SSD）與基於 LSTM 之需求預測的自主分配代理人，以實現即時感知與預測。該系統在資源受限的環境下，使用多代理人強化學習（PPO）進行去中心化協調。該方法基於智慧、異質的多代理人系統。研究結論指出，該系統提高了緊急配水的公平性、效率與韌性。

### E. Industry & Farming

### E. 工業與農業

In this subsection, studies on the industrial, agricultural, and smart-environment applications of agentic IoT are presented.

在本小節中，將介紹代理式物聯網（Agentic IoT）在工業、農業及智慧環境應用方面的研究。

Tariq et al. [39] propose an agent-based IoT framework to enable intelligent coordination and decision-making in distributed environments. They design a layered architecture that integrates sensing, communication, and agent-based processing to support adaptive system behavior. The study enables decentralized interaction and dynamic task execution among agents under heterogeneous conditions. The approach is based on intelligent agents within a multi-agent system (MAS). The study concludes that the proposed framework improves system efficiency, scalability, and coordination. Kandamali et al. [40] propose an AI-driven agricultural assistant (CottonBot) that integrates RAG and agentic AI tools to support realtime decision-making in cotton farming. They design a hybrid architecture combining domain-specific knowledge retrieval with IoT sensor data and weather APIs to generate contextaware and field-specific recommendations. The system enables dynamic irrigation decisions through agent-based tool calling and real-time data processing. The approach is based on intelligent agents within a multi-agent system (MAS). The study utilizes open-weight LLMs, particularly LLaMA 3.1 via the Ollama framework, along with models such as Mistral and Phi-3 for evaluation. Sümer et al. [41] propose a semiautonomous IoT framework introducing the concept of “smart agents as customers” for Industry 5.0 environments. They design a system integrating IoT devices, blockchain, and smart contracts to enable automated purchasing and supply management with human-in-the-loop control. The framework supports decentralized transactions and transparent tracking of interactions through Web 3.0 technologies. The approach is based on intelligent, heterogeneous multi-agent systems. The study concludes that it improves automation, transparency, and efficiency in digital commerce and smart environments. Petrović et al. [42] propose an LLM-driven approach to automate scalability and maintenance of IoT systems using model-driven engineering (MDE). They use system modeling, constraint validation (OCL), and automated code generation to support hardware integration and maintenance tasks. The framework enables automated system updates and decisionmaking through LLM-based reasoning and planning agents. It incorporates models such as GPT-4o and o1-mini for code generation and maintenance planning. The approach is based on intelligent, heterogeneous agent-based systems. The study concludes that it reduces cognitive load and improves efficiency in IoT system management. Mohanaprasad et al. [43] propose HARMONY, a multimodal LLM-driven framework for smart home IoT systems to enable context-aware and intent-driven automation. They design a multi-layered architecture where an MLLM core performs multimodal perception, intent recognition, and task planning, while an orchestrator coordinates domain-specific agents. The framework uses multimodal large language models for reasoning and orchestration. It includes models such as Qwen 2.5 VL, GPT-4.1, Gemini 2.5, and LLaMA. The approach is based on intelligent and heterogeneous multi agent systems. The study concludes that this approach improves personalization, coordination, and decision making in smart home environments.

Tariq 等人 [39] 提出了一種基於代理（agent）的物聯網框架，旨在分散式環境中實現智慧協調與決策。他們設計了一種分層架構，整合了感測、通訊與基於代理的處理，以支援自適應系統行為。該研究實現了在異質條件下代理之間的去中心化互動與動態任務執行。此方法基於多代理系統（MAS）中的智慧代理。研究結論指出，所提出的框架提升了系統效率、擴展性與協調性。Kandamali 等人 [40] 提出了一種人工智慧驅動的農業助手（CottonBot），整合了 RAG 與代理式 AI 工具，以支援棉花種植的即時決策。他們設計了一種混合架構，將特定領域的知識檢索與物聯網感測器數據及天氣 API 相結合，以生成具備情境感知且針對特定田地的建議。該系統透過基於代理的工具調用與即時數據處理，實現了動態灌溉決策。此方法基於多代理系統（MAS）中的智慧代理。該研究利用了開源權重的 LLMs，特別是透過 Ollama 框架運行的 LLaMA 3.1，並使用 Mistral 和 Phi-3 等模型進行評估。Sümer 等人 [41] 提出了一種半自主物聯網框架，為工業 5.0 環境引入了「智慧代理作為客戶」的概念。他們設計了一個整合物聯網設備、區塊鏈與智慧合約的系統，以實現具備人機協作（human-in-the-loop）控制的自動化採購與供應管理。該框架透過 Web 3.0 技術支援去中心化交易與互動的透明追蹤。此方法基於智慧、異質的多代理系統。研究結論指出，該框架提升了數位商務與智慧環境中的自動化程度、透明度與效率。Petrović 等人 [42] 提出了一種 LLM 驅動的方法，利用模型驅動工程（MDE）來自動化處理物聯網系統的擴展性與維護。他們使用系統建模、約束驗證（OCL）與自動程式碼生成來支援硬體整合與維護任務。該框架透過基於 LLM 的推理與規劃代理，實現了自動化系統更新與決策。它納入了 GPT-4o 與 o1-mini 等模型用於程式碼生成與維護規劃。此方法基於智慧、異質的代理系統。研究結論指出，該方法減輕了認知負荷並提升了物聯網系統管理的效率。Mohanaprasad 等人 [43] 提出了 HARMONY，這是一個用於智慧家庭物聯網系統的多模態 LLM 驅動框架，旨在實現情境感知與意圖驅動的自動化。他們設計了一種多層架構，其中 MLLM 核心執行多模態感知、意圖識別與任務規劃，並由協調器（orchestrator）負責協調特定領域的代理。該框架使用多模態大型語言模型進行推理與編排。 它包含 Qwen 2.5 VL、GPT-4.1、Gemini 2.5 和 LLaMA 等模型。該方法基於智慧且異質的多代理人系統。研究結論指出，此方法提升了智慧家庭環境中的個人化程度、協調能力與決策效率。

## V. Challenges and Future Research Directions

## V. 挑戰與未來研究方向

Agentic IoT has strong potential to transform traditional networks into autonomous, adaptive, and intelligent ecosystems. However, it also creates several important challenges. These challenges must be solved before agentic IoT can be deployed reliably and used in real world systems.

代理人化物聯網（Agentic IoT）具有將傳統網路轉型為自主、自適應且智慧生態系統的強大潛力。然而，它也帶來了幾項重要的挑戰。在代理人化物聯網能夠可靠地部署並應用於現實世界系統之前，必須先解決這些挑戰。

### A. Challenges A. 挑戰

Interoperability, Semantic Standardization, and Protocol Fragmentation: Interoperability, semantic standardization, and protocol fragmentation are major challenges for Agentic IoT systems. An Agentic IoT system must allow agents, devices, and services from different manufacturers to work together in the same environment. However, these components often use different data formats, APIs, and communication rules. These differences make many components incompatible with each other. As a result, each integration often becomes manual, fragile, and difficult to reuse. This also makes the system harder to scale, secure, and maintain [17]. New protocols such as MCP, A2A, and ACP have begun to offer standardized frameworks for tool access, inter-agent messaging, capability discovery, and task sharing. However, these protocols are still new and at an early stage of maturity, and they are not fully integrated with established IoT communication infrastructures [18]. In this context, the main challenge is to create a common semantic and protocol layer in which heterogeneous agents and resource-constrained devices can discover each other, declare their capabilities in a machine-readable way, share semantically consistent data, and work together without requiring a custom middleware layer for each use case.

互通性、語義標準化與協定碎片化：互通性、語義標準化與協定碎片化是代理式物聯網（Agentic IoT）系統面臨的主要挑戰。代理式物聯網系統必須允許來自不同製造商的代理、裝置與服務在同一環境中協同工作。然而，這些組件通常使用不同的資料格式、API 和通訊規則。這些差異導致許多組件彼此不相容。因此，每次整合往往變成手動、脆弱且難以重複使用。這也使得系統在擴展、安全與維護方面變得更加困難 [17]。MCP、A2A 和 ACP 等新協定已開始為工具存取、代理間訊息傳遞、能力發現和任務共享提供標準化框架。然而，這些協定仍處於起步階段，成熟度尚不足，且尚未與既有的物聯網通訊基礎設施完全整合 [18]。在此背景下，主要的挑戰在於建立一個通用的語義與協定層，使異質代理與資源受限的裝置能夠互相發現、以機器可讀的方式宣告其能力、共享語義一致的資料，並在不需要為每個案例自訂中介軟體層的情況下協同工作。

Trust, Security, and Governance: When an IoT agent not only collects data but also makes decisions and takes actions on its own, the threat space goes far beyond data privacy. New

信任、安全與治理：當物聯網代理不僅收集資料，還能自行做出決策並採取行動時，威脅空間將遠遠超出資料隱私的範疇。新的

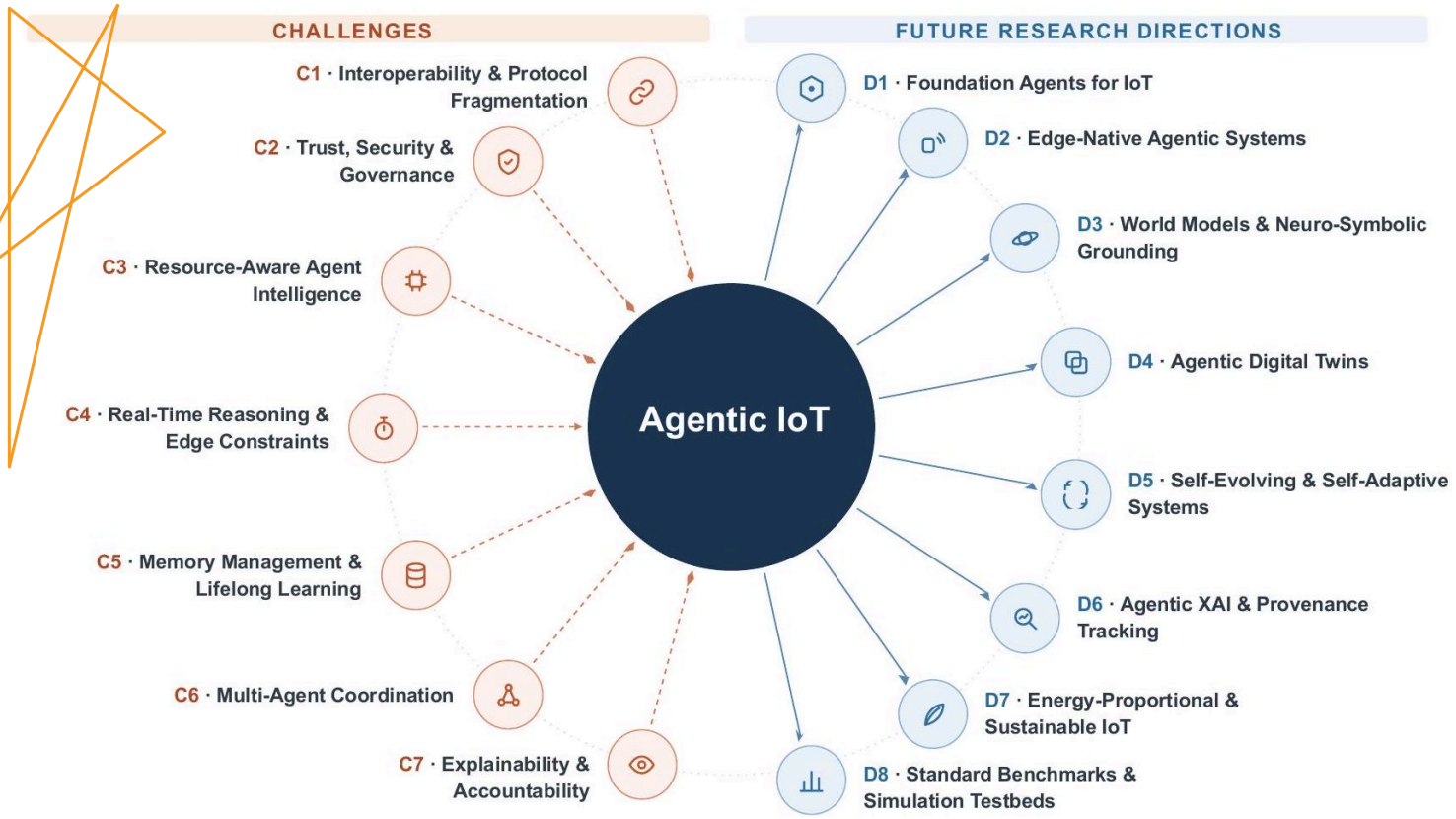


Figure 2: Fig. 2. Challenges and future research directions in Agentic IoT.

圖 2：代理式物聯網的挑戰與未來研究方向。

attack paths come to the fore, such as steering the agent's reasoning, pushing it toward unsafe tool calls, or taking over inter-agent collaboration [44]. Existing IoT security methods, however, are not designed for systems that can plan and act on their own; moreover, agent chains composed of multiple layers and manufacturers make authentication, tracing the source of a decision, and determining responsibility even more difficult [45]. Especially in safety-critical applications, how these systems will be audited remains unanswered, starting with the question of who is responsible when an autonomous decision causes harm.

攻擊路徑隨之浮現，例如引導代理的推理過程、促使其進行不安全的工具調用，或接管代理間的協作 [44]。然而，現有的物聯網安全方法並非針對能自行規劃與行動的系統而設計；此外，由多個層級和製造商組成的代理鏈，使得身分驗證、追溯決策來源以及釐清責任變得更加困難 [45]。特別是在安全至上的應用中，這些系統將如何接受稽核仍是未知數，首要問題便是當自主決策造成損害時，應由誰承擔責任。

3) Resource-Aware Agent Intelligence: Capabilities such as reasoning, planning, memory, and tool use, which make agents intelligent, are mostly fed by large language models or large-scale foundation models. However, the computing power, memory, bandwidth, and energy requirements of these models exceed the capacity of most IoT devices [46]. Although small language models are becoming increasingly powerful, adjusting model capacity according to the difficulty of the task, the risk level of the decision, and the limits of the hardware remains a challenging task [47]. The main challenge here is to design the agent from the beginning in a resource-aware, scalable, and task-sensitive manner without compromising features such as accuracy, security, context awareness, and explainability. In such an approach, the same agent architecture should be able to operate as an elastic intelligence layer that can gradually decrease and increase its capability from powerful cloud servers to edge gateways and then down to microcontroller-level devices.

3) 資源感知型代理人智慧：賦予代理人智慧的推理、規劃、記憶和工具使用等能力，大多是由大型語言模型或大規模基礎模型所驅動。然而，這些模型的運算能力、記憶體、頻寬和能源需求超出了大多數 IoT 裝置的負荷 [46]。雖然小型語言模型的功能日益強大，但根據任務難度、決策風險等級以及硬體限制來調整模型容量，仍是一項具挑戰性的任務 [47]。此處的主要挑戰在於，如何在不犧牲準確性、安全性、情境感知和可解釋性等特性的前提下，從一開始就以資源感知、可擴展且任務敏感的方式設計代理人。在這種方法中，相同的代理人架構應能作為一個彈性智慧層運作，使其能力可以從強大的雲端伺服器，逐步縮減至邊緣閘道器，再下探至微控制器等級的裝置。

4) Real-Time Reasoning and Edge Constraints: Multi-step reasoning, information retrieval, memory access, tool calling, and inter-agent coordination processes naturally create additional latency. However, in many IoT system loops, the decision must be made within milliseconds [48]. Moving reasoning to the edge device reduces the round-trip time to the server, but this time the agent has to cope with limited computing power, disconnected connections, and often incomplete or uncertain observations [49]. Therefore, the main problem is not only to run a faster model, but also to establish a risk-sensitive and latency-aware reasoning architecture that dynamically determines which decisions will be made on the local server or in the cloud.

4) 即時推理與邊緣限制：多步驟推理、資訊檢索、記憶體存取、工具調用以及代理人間的協調過程，自然會產生額外的延遲。然而，在許多 IoT 系統迴路中，決策必須在毫秒內完成 [48]。將推理移至邊緣裝置可減少往返伺服器的時間，但此時代理人必須應對有限的運算能力、斷斷續續的連線，以及通常不完整或不確定的觀測結果 [49]。因此，主要問題不僅在於運行更快的模型，還在於建立一個具備風險敏感度和延遲感知的推理架構，以動態決定哪些決策應在本地伺服器或雲端進行。

5) Memory Management and Lifelong Learning: For an agent to perform its defined operational task well, it also needs to remember the context, past interactions, the outcomes of previous decisions, and the information it has learned from the environment. However, due to reasons such as a limited context window, narrow device storage, and energy constraints, an IoT agent cannot effectively decide how much of this information it can reliably store and retrieve [50]. Therefore, the challenge is not merely to add a larger memory to the system, but to develop reliable memory management mechanisms that can decide which information will be stored, when it will be updated, when it will be forgotten, and which information will be retrieved at the moment of decision.

5) 記憶體管理與終身學習：為了讓代理人能良好地執行其定義的運作任務，它還需要記住情境、過去的互動、先前決策的結果，以及從環境中學習到的資訊。然而，由於情境視窗有限、裝置儲存空間狹小以及能源限制等原因，IoT 代理人無法有效決定這些資訊中有多少可以可靠地儲存與檢索 [50]。因此，挑戰不僅在於為系統增加更大的記憶體，還在於開發可靠的記憶體管理機制，以決定哪些資訊將被儲存、何時更新、何時遺忘，以及在決策時刻應檢索哪些資訊。

6) Multi-Agent Coordination and Collective Intelligence: In Agentic IoT, many different agents need to communicate with one another, share tasks, coordinate resources, and resolve conflicts under changing conditions and with limited information. This requirement creates the necessity for tasks to be performed by multiple agents [51]. At this point, as the number of agents increases, the communication load, coordination complexity, and risk of unexpected collective behavior also increase. Therefore, the goal in multi-agent systems is not only for each agent to perform its own local tasks in the best way, but also to produce a consistent and safe collective behavior aligned with the overall goals of the system [52]. The challenge here is to transform agents into true collective intelligence without collectively dragging the system into instability, resource competition, or unexpected errors.

6) 多代理人協調與集體智慧：在代理型物聯網（Agentic IoT）中，許多不同的代理人需要在不斷變化的條件和有限的資訊下相互溝通、分享任務、協調資源並解決衝突。這項需求使得任務必須由多個代理人共同執行 [51]。在此情況下，隨著代理人數量的增加，通訊負荷、協調複雜度以及非預期集體行為的風險也會隨之增加。因此，多代理人系統的目標不僅是讓每個代理人以最佳方式執行其局部任務，還要產生與系統整體目標一致且穩定安全的集體行為 [52]。這裡的挑戰在於如何將代理人轉化為真正的集體智慧，而不會集體導致系統陷入不穩定、資源競爭或非預期錯誤。

7) Explainability and Responsible Agentic IoT: Since agents act through a multi-step process rather than producing a single output, explaining why an IoT agent perceived, planned, and behaved in a certain way is much more difficult than interpreting the result of a fixed model [53] . In IoT environments that produce physical consequences, this uncertainty weakens not only user trust but also debugging, certification, regulatory compliance, and legal accountability. Therefore, in Agentic IoT, explainability should go beyond classical XAI methods and produce process-based explanations that cover the entire decision-making process, sensor and data provenance tracking, tool-call logs, inter-agent interaction history, and humanauditable decision traces.

7) 可解釋性與負責任的代理型物聯網：由於代理人是透過多步驟過程運作，而非僅產生單一輸出，因此解釋為何物聯網代理人以特定方式進行感知、規劃和行動，比解釋固定模型的結果要困難得多 [53]。在會產生物理後果的物聯網環境中，這種不確定性不僅削弱了使用者的信任，也影響了除錯、認證、法規遵循和法律責任歸屬。因此，在代理型物聯網中，可解釋性應超越傳統的可解釋人工智慧（XAI）方法，產生涵蓋整個決策過程、感測器與數據來源追蹤、工具調用日誌、代理人間互動歷史以及人類可稽核決策軌跡的程序化解釋。

Safety & Accounting

B. Future Research Directions

B. 未來研究方向

The emergence of Agentic IoT brings with it a broad and exciting research agenda that goes far beyond integrating intelligent agents with connected devices. Future studies need to focus on how these systems can reason under physical constraints, operate efficiently on edge devices, adapt over time, explain their decisions, and be evaluated through realistic benchmark tests. In this direction, the following topics reveal the main challenges in the maturation process of Agentic IoT and the opportunities ahead.

代理型物聯網的興起帶來了廣泛且令人興奮的研究議程，其範疇遠超將智慧代理人與連網設備整合。未來的研究需要關注這些系統如何在物理限制下進行推理、如何在邊緣設備上高效運作、如何隨時間演進進行適應、如何解釋其決策，以及如何透過現實的基準測試進行評估。依此方向，以下主題揭示了代理型物聯網成熟過程中的主要挑戰以及未來的機遇。

Foundation Agents for IoT: Agent research is converging on the concept of a foundation agent, whose cognitive core is a foundation model and which is enhanced with modular memory, planning, tool use, and perception [54]. However, these general-purpose agents have not been grounded in the physical constraints of IoT, while IoT-oriented foundation models have remained narrow and task-specific perception backbones lacking autonomous reasoning [55]. Therefore, future studies should develop domain-specific foundation agents that integrate a general reasoning core with SLM models and local perception backbones suitable for the nature of IoT; thus, a single agent can generalize across different devices and tasks with minimal labeled data and on-device efficiency.

Protocol Role : Base IoT Agents

IoT 基礎代理人（Foundation Agents for IoT）：代理人研究正趨向於「基礎代理人」的概念，其認知核心為基礎模型，並透過模組化記憶、規劃、工具使用與感知進行增強 [54]。然而，這些通用型代理人尚未與 IoT 的物理限制接軌，而面向 IoT 的基礎模型仍侷限於特定任務的感知骨幹，缺乏自主推理能力 [55]。因此，未來的研究應開發領域特定的基礎代理人，將通用推理核心與適合 IoT 特性的小型語言模型（SLM）及本地感知骨幹相結合；如此一來，單一代理人便能以極少的標籤數據與裝置端效率，在不同設備與任務間實現泛化。

Edge-Native Agentic Systems: Studies on Edge General Intelligence propose running agentic AI directly on edge devices and distributing capabilities such as perception, reasoning, and planning across the device-edge-cloud continuum [56]. Most existing agents, however, still assume cloud-scale computing power and uninterrupted connectivity. Therefore, current systems are not compatible with the latency, energy, and intermittent connectivity realities of IoT. Accordingly, future studies need to develop edge-native agent designs that combine lightweight and quantized agent models with energyaware planning. In this way, the perception-reasoning-action loop can operate on the device itself even under strict resource constraints.

Local agent

邊緣原生代理系統（Edge-Native Agentic Systems）：邊緣通用智能（Edge General Intelligence）的研究提議直接在邊緣設備上執行代理式 AI，並將感知、推理與規劃等能力分布於「設備—邊緣—雲端」連續體中 [56]。然而，現有的大多數代理人仍預設具備雲端級的運算能力與不間斷的連線。因此，目前的系統無法相容於 IoT 環境中延遲、能源限制及間歇性連線的現實。據此，未來的研究需要開發邊緣原生的代理人設計，將輕量化與量化後的代理模型與能源感知規劃相結合。透過這種方式，「感知—推理—行動」迴路即使在嚴格的資源限制下，也能在設備本身運行。

World Models and Neuro-Symbolic Physical Grounding: One of the strongest research directions in Agentic IoT systems is to move the agent beyond a structure that merely reacts to current sensor data and support it with a predictive world model on which it can reason, plan, and test possible outcomes in advance [57]. Indeed, it is clear that general-purpose agents need world models that represent the causal and dynamic structure of their environments in order to make safe and consistent decisions. The literature on physical simulators, embedded intelligence, and embodied intelligence also shows that such models play a critical role in reducing the sim-toreal gap, generating counterfactual scenarios, and supporting

safe planning processes [58]. In the context of Agentic IoT, this approach refers to a transition toward explainable and autonomous IoT architectures that integrate the sensor-motor loop with a neuro-symbolic reasoning core and reduce the risk of hallucination by constraining the agent's actions with physical laws, system constraints, and symbolic rules.

世界模型與神經符號物理接地（World Models and Neuro-Symbolic Physical Grounding）：代理式 IoT 系統中最強大的研究方向之一，是讓代理人超越僅對當前感測器數據做出反應的結構，轉而支持具備預測性的世界模型，使其能預先進行推理、規劃並測試可能的結果 [57]。事實上，通用型代理人顯然需要能代表其環境因果與動態結構的世界模型，以做出安全且一致的決策。關於物理模擬器、嵌入式智能與具身智能（Embodied Intelligence）的文獻也顯示，此類模型在縮小「模擬至現實」（sim-to-real）差距、生成反事實情境以及支持安全規劃流程方面發揮著關鍵作用 [58]。在代理式 IoT 的背景下，此方法意指轉向可解釋且自主的 IoT 架構，將感測器—馬達迴路與神經符號推理核心整合，並透過物理定律、系統約束與符號規則來限制代理人的行動，進而降低幻覺風險。

Agentic Digital Twins: Digital twins are evolving from passive copies that merely reflect the system into "agentic digital twin" structures that actively guide and reshape it [59]. On the IoT side, twins have so far mostly remained limited to monitoring and "what-if" scenarios, and they have not had the ability to act on their own or update their models autonomously. However, when an autonomous agent is embedded into the twin, the twin can continuously monitor its physical counterpart, intervene in it, and complete the perception-action loop at its own level [60]. As the twin's power to affect the system it models increases, mechanisms will be needed to control risks such as performative lock-in and self-confirming model drift.

代理式數位分身（Agentic Digital Twins）：數位分身正從僅反映系統的被動副本，演變為能主動引導並重塑系統的「代理式數位分身」結構 [59]。在物聯網（IoT）方面，分身至今大多仍侷限於監控和「假設分析」（what-if）情境，尚不具備自主行動或自主更新模型的能力。然而，當自主代理被嵌入分身中，分身便能持續監控其實體對應物、對其進行干預，並在自身層級完成「感知-行動」循環 [60]。隨著分身影響其建模系統的能力增強，將需要相關機制來控制諸如「執行性鎖定」（performative lock-in）和「自我確認模型漂移」等風險。

Self-Evolving and Self-Adaptive Systems: A rapidly growing research area is self-evolving agents that, instead of remaining unchanged after deployment, can improve their model, memory, tools, and even architecture by learning from experience [61]. Today's Agentic IoT systems, however, are mostly based on static models or models that are retrained at certain intervals. The performance of these models decreases as conditions and sensor data change. Future research should aim to enable agents to improve themselves throughout their entire operational lifetime by bringing together lifelong learning, retrieval-augmented memory, and online adaptation. This autonomy, however, needs to be balanced with appropriate safeguards against the risk of misevolution.

自我演化與自我適應系統：一個快速增長的研究領域是自我演化代理，它們在部署後並非保持不變，而是能透過經驗學習來改進其模型、記憶、工具甚至架構 [61]。然而，現今的代理式物聯網系統大多基於靜態模型，或以特定間隔進行重新訓練的模型。隨著環境條件和感測器數據的變化，這些模型的性能會隨之下降。未來的研究應旨在結合終身學習、檢索增強記憶（RAG）和在線適應，使代理能在整個運作生命週期中不斷自我提升。然而，這種自主性需要在防止「錯誤演化」風險的適當保護措施之間取得平衡。

Agentic Explainable AI and Provenance Tracking: Since an agent acts through a multi-step process involving tool use, memory, and coordination rather than producing a single output, what needs to be explained is no longer only a single result but the entire process itself. In Agentic IoT, where agents take physical actions that produce real-world consequences, reasoning that cannot be understood damages both trust and accountability. XAI methods in the literature, however, have mostly been designed for fixed perception models [62]. Therefore, future studies should develop agentic XAI models adapted to IoT. Ensuring accountability at every step of the process, providing provenance information that shows which sensor a decision is based on, and creating records through which humans can trace why an agent perceived and acted in a certain way are becoming highly important for explainability.

代理式可解釋人工智慧與溯源追蹤：由於代理是透過涉及工具使用、記憶和協調的多步驟過程來行動，而非僅產生單一輸出，因此需要解釋的不再只是單一結果，而是整個過程本身。在代理式物聯網中，代理採取的實體行動會產生現實世界的後果，無法被理解的推理過程會損害信任與問責制。然而，文獻中的可解釋人工智慧（XAI）方法大多是為固定感知模型所設計 [62]。因此，未來的研究應開發適用於物聯網的代理式 XAI 模型。確保過程中的每一步都具備問責性、提供顯示決策依據感測器的溯源資訊，並建立讓人類能追蹤代理為何以特定方式感知與行動的記錄，對於可解釋性而言正變得至關重要。

Energy-Proportional and Sustainable Agentic IoT: The real-world scalability of Agentic IoT systems requires energy to be treated not merely as a secondary optimization metric, but as a central design constraint. Especially for agents operating on resource-constrained edge devices, quantization, pruning, early-exit mechanisms, hardware-aware inference, and networking-aware planning approaches are critically important for sustainably balancing accuracy, latency, computational cost, and energy consumption [63]. In this context, the future direction is evolving toward energy-proportional agent architectures that dynamically adjust their workload according to the complexity of the physical task, network conditions, and the risk level of the decision, rather than agents that operate continuously at fixed capacity. In such an approach, the main goal is not only to produce more accurate decisions, but also to make Agentic IoT systems long-lived and low-carbon-

footprint systems by reducing the energy cost per decision.

能源比例與永續的代理式物聯網：代理式物聯網系統在現實世界的擴展性，要求將能源視為核心設計約束，而不僅僅是次要的優化指標。特別是對於在資源受限的邊緣設備上運行的代理，量化、剪枝、提前退出機制、硬體感知推理以及網路感知規劃方法，對於永續平衡準確性、延遲、運算成本和能耗至關重要 [63]。在此背景下，未來的發展方向正朝著能源比例代理架構演進，這種架構會根據物理任務的複雜程度、網路狀況和決策風險等級動態調整其工作負載，而非以固定容量持續運行。在這種方法中，主要目標不僅是產生更準確的決策，還包括透過降低每次決策的能源成本，使代理式物聯網系統成為長效且低碳足跡的系統。

Standard Benchmarks and Simulation Testbeds: For the field to mature, common testbeds are needed that evaluate agentic IoT systems in a realistic and reproducible way. Recent taxonomies on agent evaluation are moving toward measuring trajectory-level behavior, reliability, and safety [64]. Future studies should complement these with high-fidelity simulation environments and operational metrics that enrich them along the real-time, energy, and physical-consequence dimensions of IoT. Such an infrastructure would enable different approaches to be compared on a common ground and help avoid the illusion of progress.

標準基準測試與模擬測試平台：為了使該領域走向成熟，需要通用的測試平台，以現實且可重複的方式評估代理式物聯網系統。近期關於代理評估的分類法正轉向衡量軌跡級行為、可靠性與安全性 [64]。未來的研究應輔以高保真模擬環境和運作指標，並沿著物聯網的即時性、能源和物理後果維度來豐富這些指標。這樣的基礎設施將使不同的方法能在共同的基礎上進行比較，並有助於避免進步的錯覺。

## VI. Conclusion VI. 結論

In this paper, we examined the emerging Agentic IoT paradigm and took a first step toward establishing it as a systematic research field. We formally defined the concept, positioned it in relation to IoT, AIoT, and the Internet of Agents, and traced the paradigm shift from rule-based and model-based systems toward autonomous, agent-centric ecosystems. We then introduced a three-tier reference architecture with a cross-layer agentic intelligence plane, described the cognitive loop through which agents perceive, reason, plan, act, and learn, and explained how memory, retrieval-augmented knowledge, tool use, and protocol bridging make this loop operational across the device-edge/fog-cloud continuum. Our literature review also reveals a clear trajectory. Early Internet of Agents studies were mainly grounded in symbolic, rule-based, and ontology-driven agent architectures, whereas recent work increasingly relies on LLM-driven reasoning, agentic tool use, and dedicated agent protocols, with applications now spreading across smart cities, disaster response, network management, agriculture, and industry. The research directions presented in this paper, from foundation agents and edge-native designs to agentic digital twins and standardized benchmarks, show where progress is most needed. We hope this study provides a common ground for researchers and practitioners working on these problems and positions Agentic IoT as the practical path through which the Internet of Agents vision reaches the physical world.

在本論文中，我們探討了新興的代理式物聯網（Agentic IoT）範式，並為將其建立為系統性研究領域邁出了第一步。我們正式定義了此概念，定位了其與物聯網（IoT）、人工智慧物聯網（AIoT）及代理人網路（Internet of Agents）的關係，並追溯了從基於規則與模型的系統轉向以代理人為中心的自主生態系統之範式轉移。隨後，我們提出了一個具備跨層級代理智慧平面的三層式參考架構，描述了代理人進行感知、推理、規劃、行動與學習的認知迴圈，並解釋了記憶、檢索增強知識、工具使用及協定橋接如何使該迴圈在裝置-邊緣/霧-雲端連續體中運作。我們的文獻回顧也揭示了一條清晰的發展軌跡：早期的代理人網路研究主要基於符號化、規則化及本體驅動的代理人架構，而近期的研究則日益依賴 LLM 驅動的推理、代理工具使用及專用代理協定，其應用現已擴展至智慧城市、災難應變、網路管理、農業及工業領域。本文提出的研究方向，從基礎代理人與邊緣原生設計，到代理式數位分身及標準化基準測試，指出了最需要取得進展的領域。我們希望這項研究能為致力於解決這些問題的研究人員與從業者提供共同基礎，並將代理式物聯網定位為實現代理人網路願景並落地於物理世界的實踐路徑。

## Declaration on Generative AI

## 生成式 AI 聲明

During the preparation of this work, the authors used ChatGPT and Claude in order to: Grammar and spelling check. After using these services, the authors reviewed and edited the content as needed and take full responsibility for the publication's content.

在準備本作品的過程中，作者使用了 ChatGPT 與 Claude 以進行：語法與拼字檢查。在使用這些服務後，作者根據需要對內容進行了審閱與編輯，並對出版內容承擔全部責任。

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